

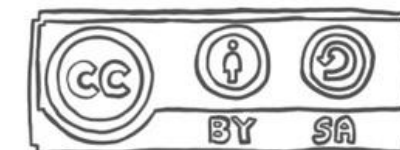
E-Textiles and Functional Materials

slides by Dr. Marion Koelle
presented by Anika Bork

inf175

University of Oldenburg, Summer Term 2024

Author's Note: unless indicated otherwise, all sketched figures and icons are hand-drawn by myself (MKoe, 03/2024) and licensed as Creative Commons.



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1. **The Musical Jacket (1998)**

2. www.maggieorth.com/art_Jacket.htm

3. Orth, M., Smith, J. R., Post, E. R., Strickon, J. A., & Cooper, E. B. (1998). Musical Jacket. In ACM SIGGRAPH 98 Electronic Art and Animation Catalog (p. 38); online at <https://digitalartarchive.siggraph.org/artwork/maggie-orth-j-r-smith-e-r-post-j-a-strickon-e-b-cooper-musical-jacket/>

4. Video: Joe Paradiso, <https://paradiso.media.mit.edu/SpectrumWeb/captions/Jackets.html>

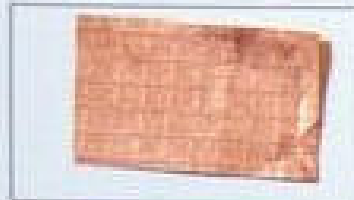




MANUFACTURING INTERACTIVE DENIM

CONDUCTIVE MATERIALS

KOBAKANT, 2017



Copper Conductive Fabric

Corrosion proof copper-silver plated polyamide ripstop fabric.
Producer: Statex
Distributor: LessEMF



Pure Copper Polyester Taffeta

Pure copper, tarnish resistant finish. High conductivity, resistivity: 0.05 Ohm/sq.
Distributor: LessEMF



VeilShield

Woven 132/inch mesh polyester fibers coated with Zinc-blackened Nickel over Copper for better corrosion resistance. Nickel allergy! 70% light transmission. 0.1 Ohm/sq. Distributor: LessEMF



SaniSilver

Double side weave, one side is highly conductive (<1 Ohm per sq) pure Silver, the other side is pure cotton (~ 100 Ohm/sq)
Producer: Statex
Distributor: LessEMF



Eeontex Non-Woven

Carbon doped Polyester/Nylon with a conductive polymer formulation.
Producer: Eeonyx
Distributor: Sparkfun



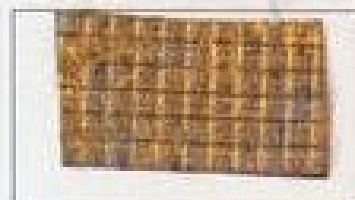
Eeontex Stretch Fabric

A conductive knitted nylon/elastane fabric with a conductive polymer formulation. Stretch in both direction.
Producer: Eeonyx
Distributor: Sparkfun



Eeontex Twill Fabric

Woven polyester fabric coated with a conductive polymer formulation.
Producer: Eeonyx



Ripstop Silver Fabric

Pure Silver coated onto nylon RipStop. Comfortable and safe against the skin. Resistivity is less than 0.25 Ohm/sq.
Producer: Statex
Distributor: LessEMF



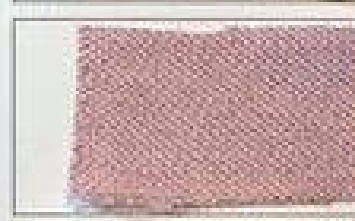
Silver Stretch Conductive Fabric

Silver plated knitted fabric, 78% Polyamide + 22% Elastomer plated with 99% pure silver.
Producer: Statex
Distributor: LessEMF



Safety Silk

Natural Silk plus Pure Silver. <1 Ohm/sq.
Distributor: LessEMF



Soft&Safe Shielding Fabric

70% bamboo fiber and 30% Silver. high conductivity (<1 Ohm per sq).
Distributor: LessEMF



Conductive Wool

Wool 80% stainless steel fiber 20%. Resistive material. Suitable for felting.
Producer: Bekaert



Velostat

Piezoresistive carbon impregnated plastic sheet material.
Producer: 3M
Distributor: LessEMF



ESD Foam

Anti-static foam used for packaging material.

Karl-Grimm High Flex 3981

Fine copper fiber plied with synthetic fiber core. Foldable.
Producer: Karl Grimm | Distributer: Karl Grimm

Karl-Grimm High Flex 3981 Silver 14/000

Fine copper fiber coated with silver plied with synthetic fiber core. Foldable. Producer: Karl Grimm | Distributer: Karl Grimm

Elitex

235/34 Polyamide plated with silver.
Producer: Imbut GmbH | Distributer: Imbut GmbH

Shieldex

Silver Plated synthetic thread.
Producer: Statex | Distributer: Statex

Bekinox VN

Fine stainless steel fiber plied. Good for heating use.
Producer: Bekaert | Distributer: Bekaert

25% Metal Egypt Color Gold Gimp

Thread wrapped with metal wire.
Producer: Bart and Fransis | Distributer: Bart and Fransis

Radio Conductive Thread

90% polyester SuperTech + 10% 0.035mm Inox Steel thread
Producer: Bart and Fransis | Distributer: Bart and Fransis

Silver Plated Nylon

66 Yarn 22+3ply 110 PET. Resistance: <1000 Ohm/10cm.
LessEMF

Nm10/3 Conductive Yarn

80% polyester 20% stainless steel, light grey.
Producer: Bekaert | Distributer: Bekaert

Silverspun Yarn

87% combed cotton, 5% silver, 5% nylon, 3% Spandex. Highly conductive, about 10 Ohm per inch. The silver will not wash out. LessEMF

Learning Goals

1. After this lecture you can ...
2. name common types of conductive (e-textile) materials.
3. interpret data sheets describing conductive (e-textile) materials.
4. determine whether a given material is electrically conductive
5. determine the resistivity of a given material.
6. make an informed selection of an e-textile material to match your project's requirements.

Legend:



Recommended reading



References



Solo Exercise

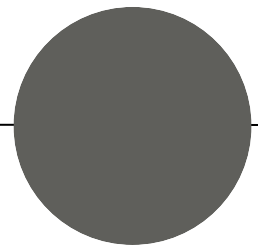


Group Exercise /
Discussion



Hands-on Exercise

E-Textile Sensors and other Components



Contact Sensors (Digital)



Fabric Button
<http://www.kobakant.at/DIY/?p=48>



Tilt Sensor
<http://www.kobakant.at/DIY/?p=201>

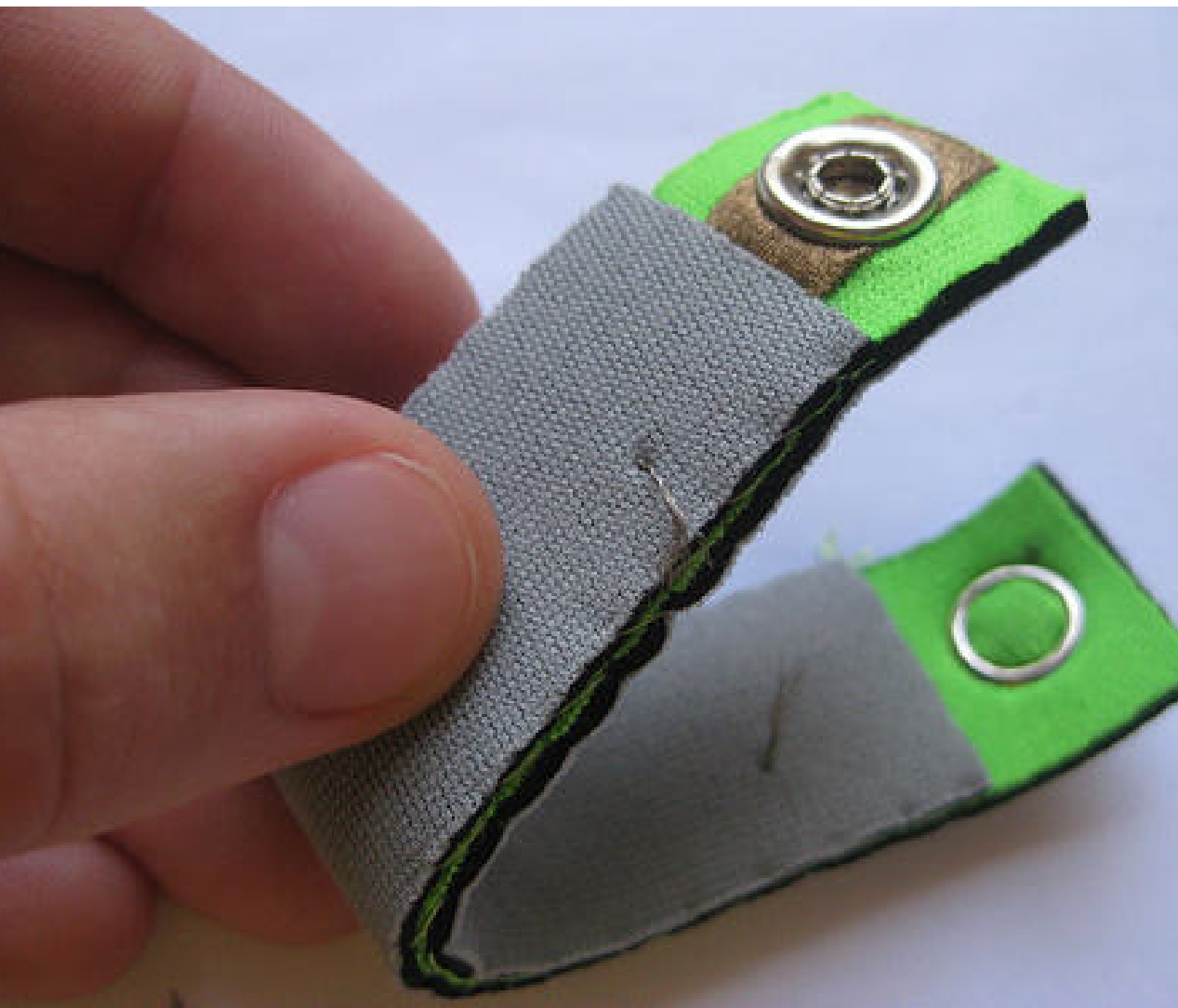


Neoprene stroke bracelet
<http://www.kobakant.at/DIY/?p=5807>

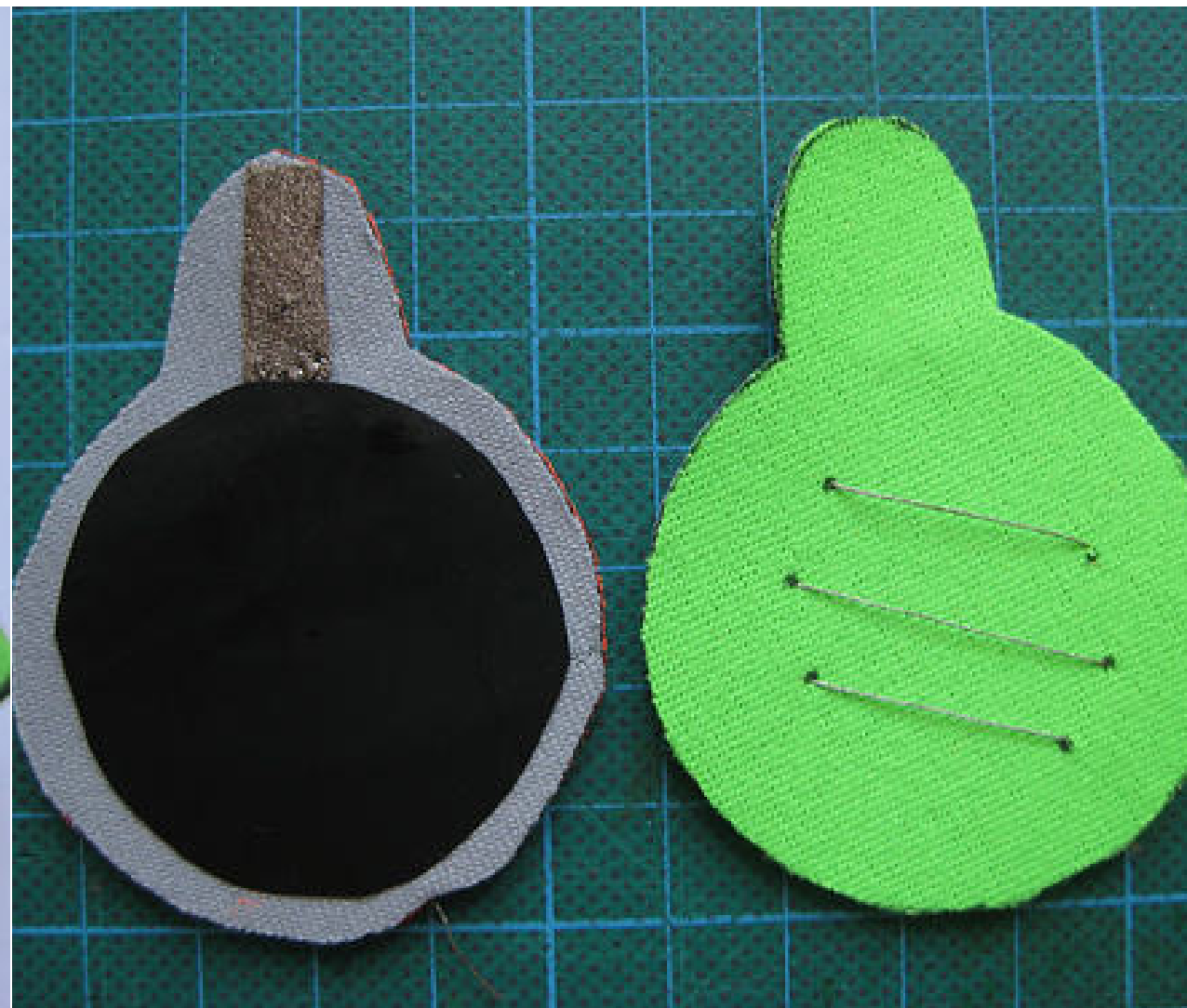


1. Hannah Perner-Wilson and Mika Satomi: Kobakant, online a <https://www.kobakant.at> (retrieved 04/2024)

Stretch and Pressure Sensors (Analog)



Neoprene Bend Sensor
<http://www.kobakant.at/DIY/?p=20>



Neoprene Pressure Sensor
<http://www.kobakant.at/DIY/?p=65>



Knit Stretch Sensor
<http://www.kobakant.at/DIY/?p=2108>



1. Hannah Perner-Wilson and Mika Satomi: Kobakant, online a <https://www.kobakant.at> (retrieved 04/2024)

Textile Microcontroller Boards



Fabric Breakout Board for Teensy
<https://www.kobakant.at/DIY/?p=2672>



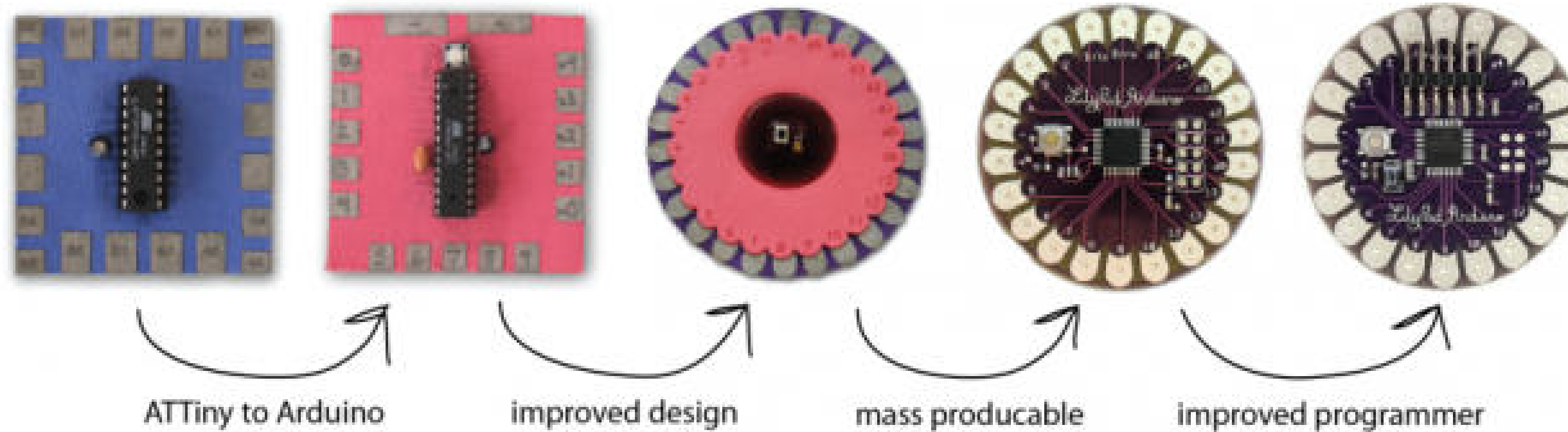
Textile Lilypad Board, Leah Buechley
Makezine (2014)



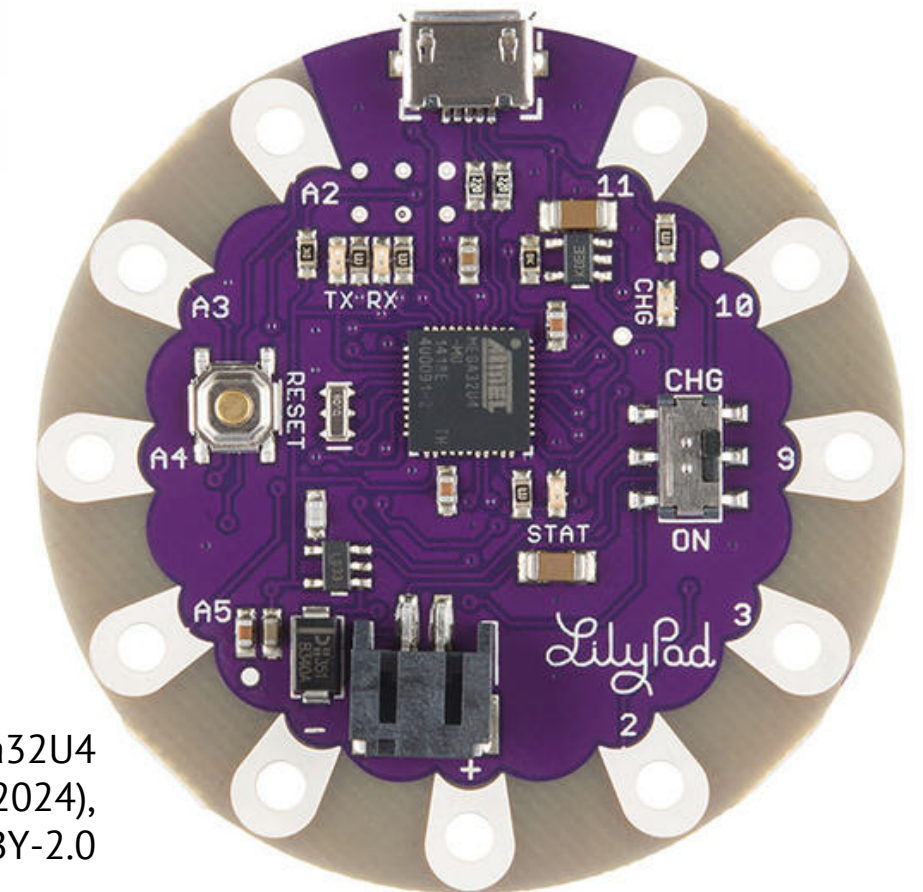
Embroidered Lilypad Circuit, Becky Stern
Wikimedia Commons, CC-BY-SA 2.0

1. Hannah Perner-Wilson and Mika Satomi: Kobakant, online at <https://www.kobakant.at> (retrieved 04/2024)
Leah Buechley: Crafting the Lilypad Arduino (2014), online at <https://makezine.com/article/maker-news/leah-buechley-crafting-the-lilypad-arduino/> (retrieved 04/2024)

Arduino LilyPad



Textile Lilypad Board, Leah Buechley
Makezine (2014)



Arduino LilyPad USB – ATmega32U4
Board (produced by Sparkfun, 2024),
Image by Sparkfun, CC-BY-2.0

CHI 2008 Proceedings · Aesthetics, Awareness, and Sketching

April 5-10, 2008 · Florence, Italy

The LilyPad Arduino: Using Computational Textiles to Investigate Engagement, Aesthetics, and Diversity in Computer Science Education

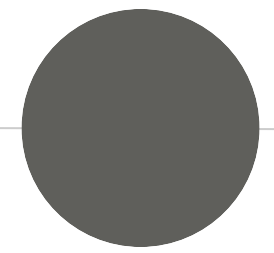
Leah Buechley, Mike Eisenberg, Jaime Catchen, and Ali Crockett
Craft Technology Group, Department of Computer Science
University of Colorado at Boulder
Boulder, CO 80309 USA
leah.buechley, duck, jaime.catchen, @colorado.edu, alicrockett@gmail.com

ABSTRACT

The advent of novel materials (such as conductive fibers) combined with accessible embedded computing platforms have made it possible to re-imagine the landscapes of fabric and electronic crafts—extending these landscapes with the creative range of electronic/computational textiles.

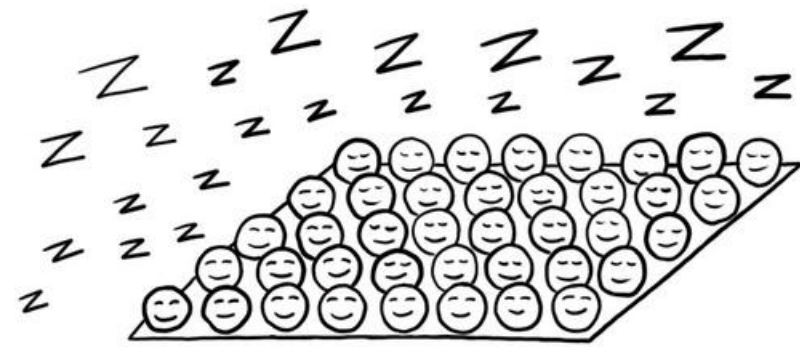
expanded portrait of “computing” more generally. People no longer interact with desktop applications exclusively, but with a burgeoning landscape of computational media—including handheld computers, “intelligent” or adaptive sensor-equipped environments, and robotic pets, just to name a few. Human computer interaction takes place

1. Leah Buechley: Crafting the Lilypad Arduino (2014), online at <https://makezine.com/article/maker-news/leah-buechley-crafting-the-lilypad-arduino/> (retrieved 04/2024)
Buechley, L., Eisenberg, M., Catchen, J., & Crockett, A. (2008, April). The LilyPad Arduino: Using Computational Textiles to Investigate Engagement, Aesthetics, and Diversity in Computer Science Education. In Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (pp. 423-432).

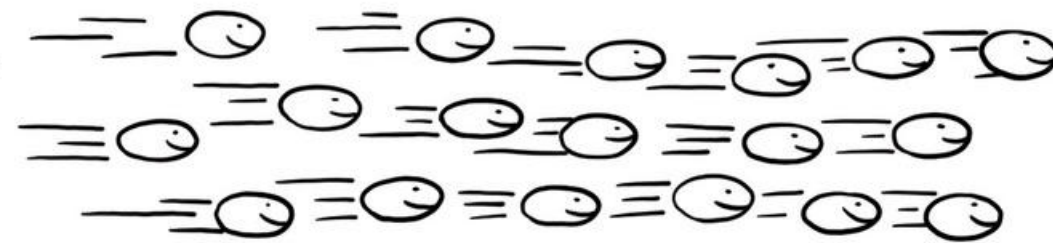


Conductivity & Resistivity

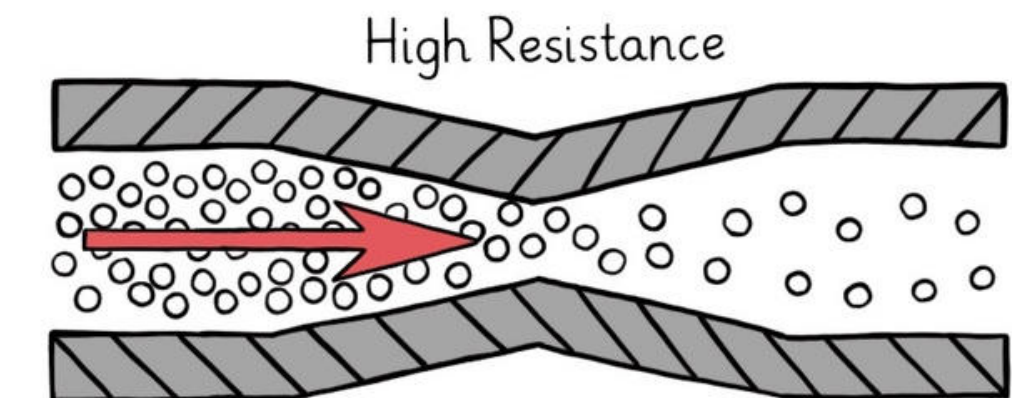
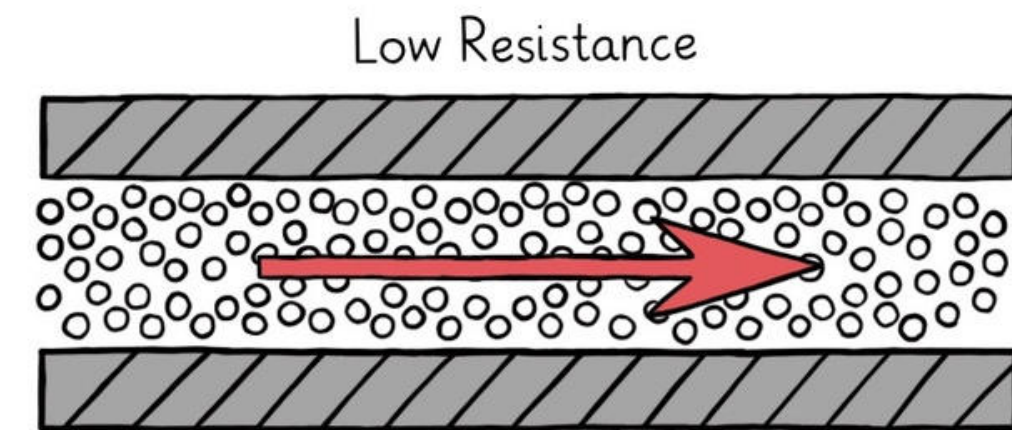
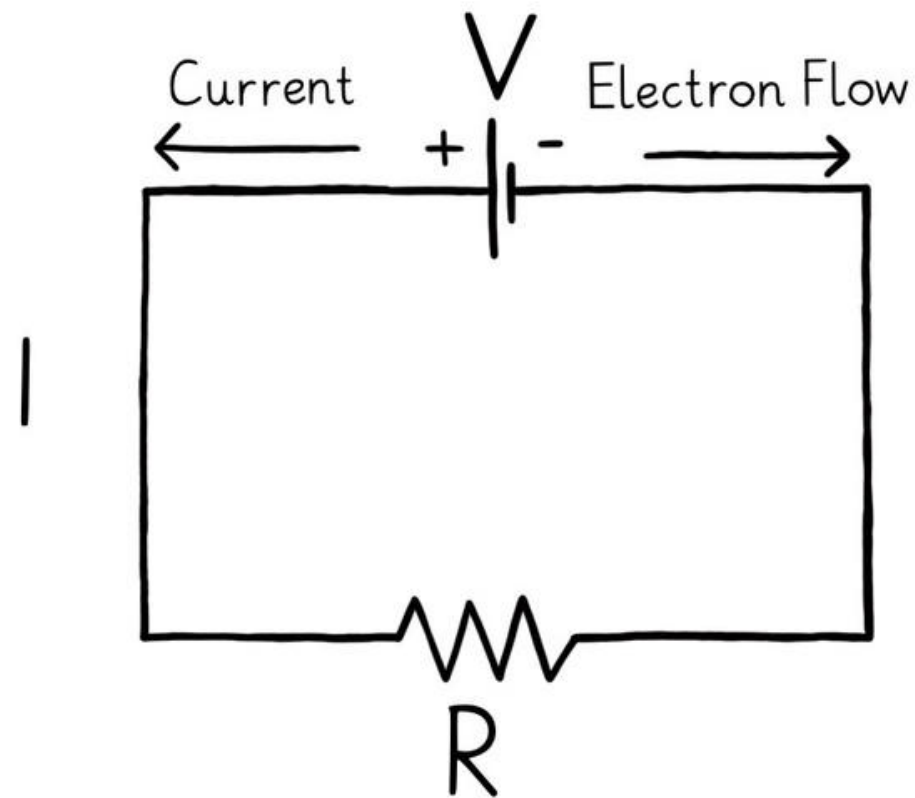
Electrons & Conductivity



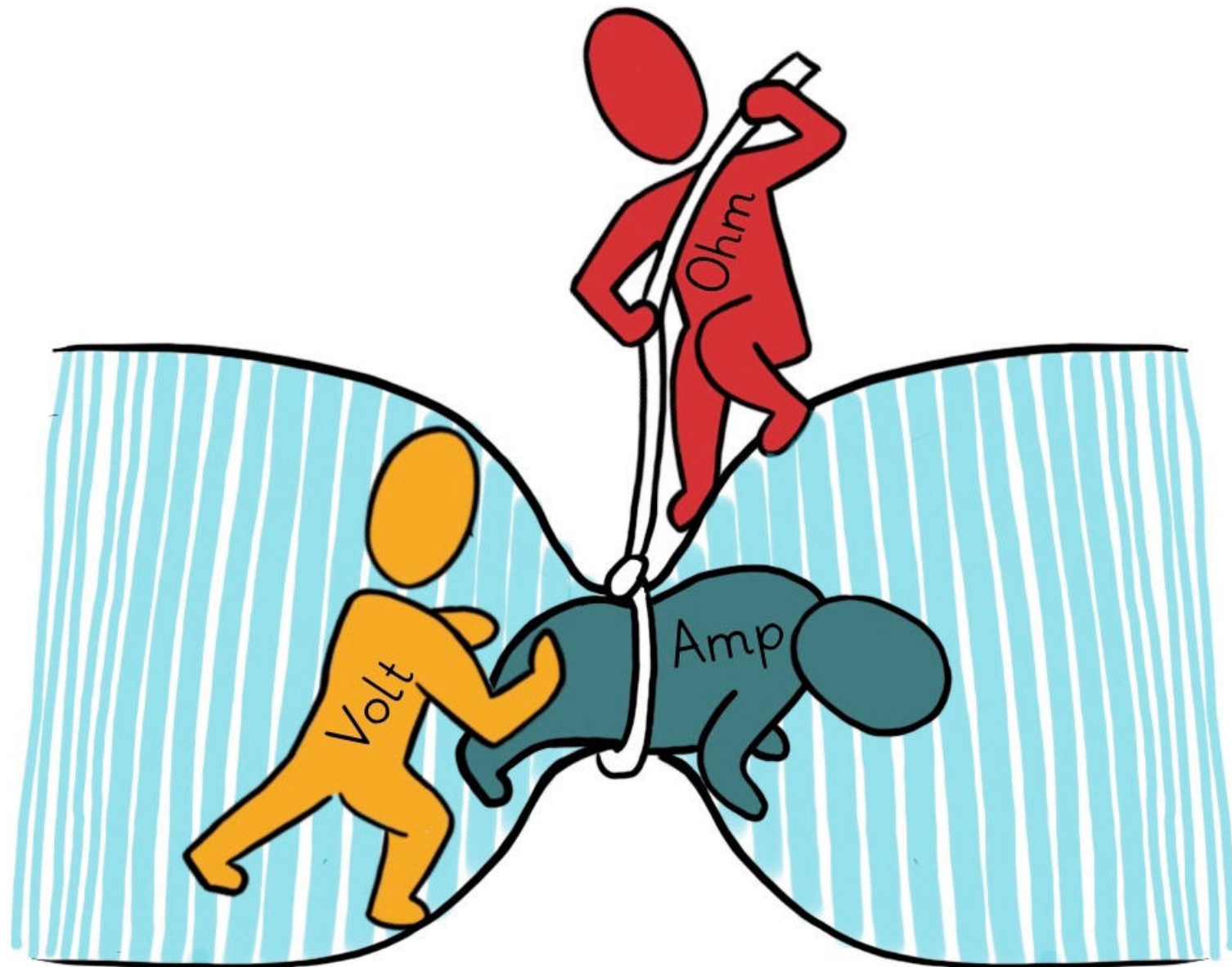
Resting Electrons



Moving Electrons



Ohm's Law



1. = Voltage (like a battery, expressed in Volt)
2. = Current (electron flow, expressed in Ampere, A colloquially also “amps”)
3. = Resistance (extend to which the material or component resists electron flow, expressed in ohms, Ω)



Using a Multimeter

1. Different measurement modes:
2. Resistance
3. Current
4. Voltage
5. Crucial tool to diagnose circuits, testing batteries, and determining a materials conductivity.

1. Tutorial: How to use a Multimeter, Sparkfun (2015), <https://learn.sparkfun.com/tutorials/how-to-use-a-multimeter/>
2. Image Source: Voelkner, Voltcraft MT-52-SE

Reading Data Sheets – Conductivity

Ag Cu Ni Sn	
Shieldex® 235/36 4-ply HC + B 200121235364HCB	
Material	Polyamide / Nylon 6.6
Metal Plated	99.9 % Pure Silver
Coating	Nitrile rubber as additional protective coating
Yarn count raw	235/36 - 2/+4 dtex
Yarn count silverized	1.220 ± 15 dtex
Twist	Z 400
Electrical Resistivity	40 Ω/m ± 10 Ω/m
Yield	8.060 m/kg ± 2.5 %
Tenacity	47 ± 5 cN/tex
Elongation / Break	26 % ± 6
Make Up	On cones 230 mm 3" 30
Packaging	Average 820 g net yarn per cone 12 or 24 cones per carton
Storage and Handling	According to our care and handling instructions
Compliance and Certification	DIN EN ISO 9001:2015, REACH, RoHS
Attention: Revised 01.06.2021 (H) - This document has been compiled from our manufacturing data according to the latest state of development, production and testing procedures. The data is subject to change without notice. The data is subject to change without notice. The data is subject to change without notice. The data is subject to change without notice.	
Sales Production and Vertrieb GmbH Werner-Str. 1-3 20357 Bremen, Germany	
Contact: Tel: +49 421 27 50 47 Fax: +49 421 27 36 43 info@shields.de	

Ag Cu Ni Sn	
Shieldex® Kassel RS 1300101130	
Raw Material	100 % Polyamide / Nylon 6.6
Weave	Ripstop
Thread density warp	400-500 threads/dm
Thread density weft	450-490 threads/dm
Metal Plated	99.9 % Pure Silver + Copper
Metal Content	8.60 ± 2 % Silver + 49.22 ± 3 % Copper + 4.55 ± 1 % CR Coating
Electrical Surface Resistivity	≤ 0.03 Ω/□
Measured Frequency	0.2 GHz - 14 GHz
Shielding Effectiveness 1	Average of < 88 dB from 0.2 GHz - 2 GHz
Shielding Effectiveness 2	Average of < 75 dB from 2 GHz - 5 GHz
Shielding Effectiveness 3	Average of < 70 dB from 5 GHz - 14 GHz
Total Weight	85 g/m² ± 12 %
Total Thickness	0.11 mm ± 10 %
Roll Width	131 ± 4 cm
Roll Length	Average 200 m
Temperature Range	-30 °C to 90 °C
Storage and Handling	According to our care and handling instructions
Compliance and Certification	DIN EN ISO 9001:2015, REACH, RoHS
Attention: Revised 01.06.2021 (H) - This document has been compiled from our manufacturing data according to the latest state of development and production technology. The data is subject to change without notice. The data is subject to change without notice. The data is subject to change without notice. The data is subject to change without notice.	
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Contact: Tel: +49 421 27 50 47 Fax: +49 421 27 36 43 info@shields.de	

Yarn count raw	235/36 - 2/+4 dtex
Yarn count silverized	1.220 ± 15 dtex
Twist	Z 400
Electrical Resistivity	40 Ω/m ± 10 Ω/m

Metal Plated	99.9 % Pure Silver + Copper
Metal Content	8.60 ± 2 % Silver + 49.22 ± 3 % Copper + 4.55 ± 1 % CR Coating
Electrical Surface Resistivity	≤ 0.03 Ω/□

Pay attention!: for e-textile materials, resistivity is typically not given in ohms (Ω).

- Volume resistivity: Ω/m (ohm-meter), used for conductive thread or yarn.
- Sheet resistivity: Ω/□ (ohm-square), used for conductive fabric, films or traces.

Volume Resistivity (Ω/cm or Ω/m)

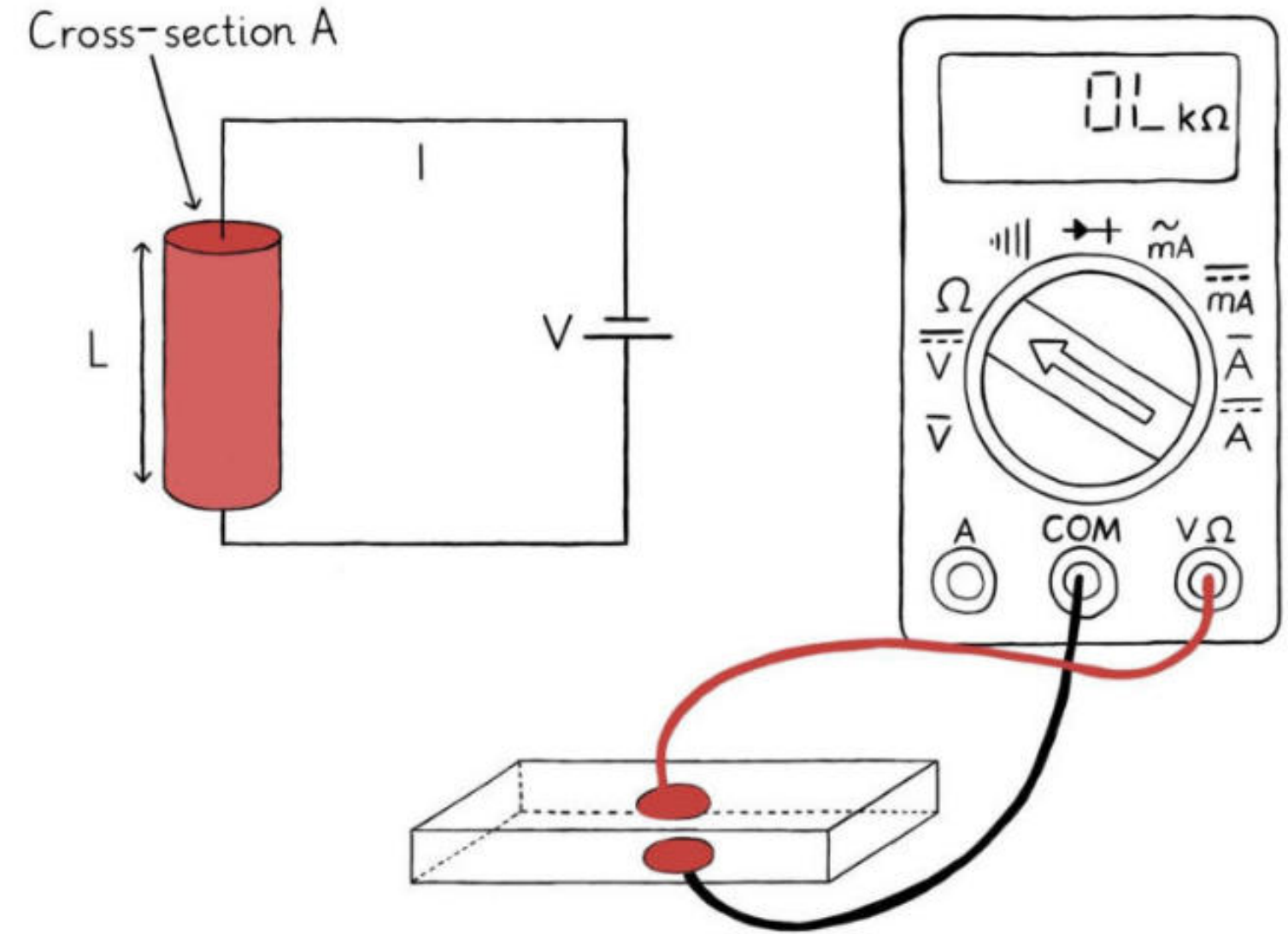
$$\rho = R * A/L$$

ρ = Volume Resistivity (Ω/cm)

R = Resistance (Ω)

A = Cross-sectional area of the material (cm^2), can be diameter (e.g., of a yarn) or width*thickness (e.g., of a foam).

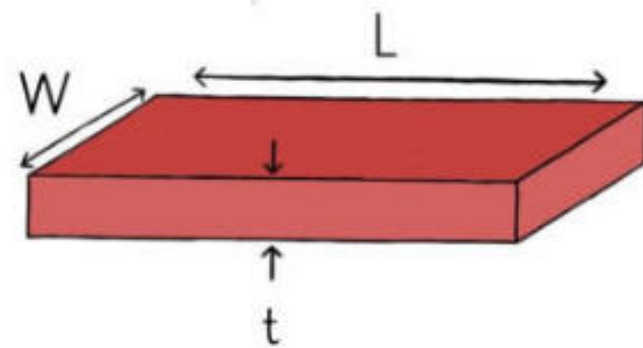
L = Distance between the probe electrodes (cm).



Volume resistivity (or bulk resistivity) is used to measure how strongly a material resists the flow of electric current. It is expressed in Ω/cm .

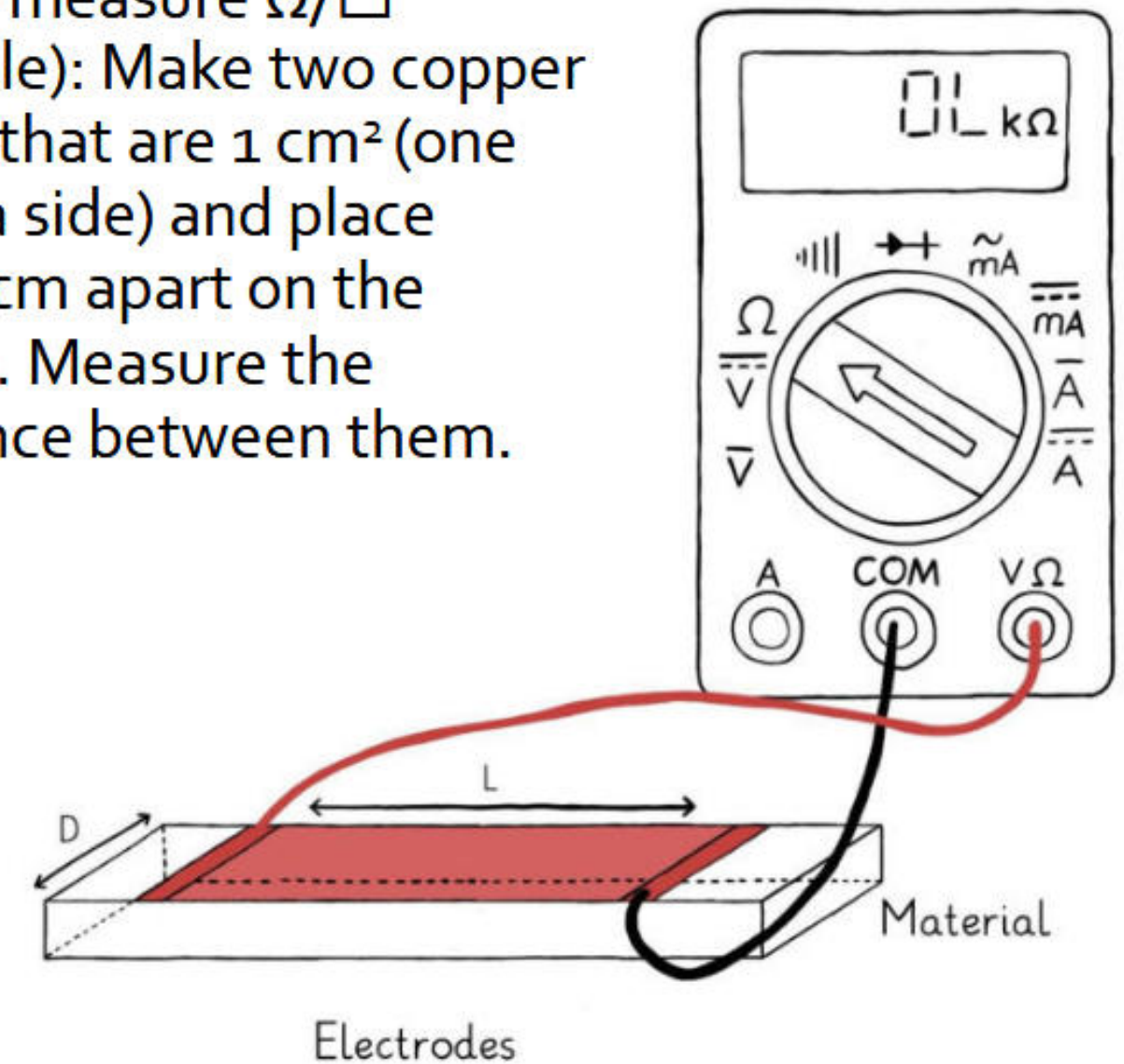
Surface Resistivity (Ω/sq or $\Omega/\square$)

Surface resistivity (also sheet resistivity) describes the resistance of a **square piece** of a **thin** material with contacts made to two opposite sides of the square.



Applies to materials where thickness t is negligible, e.g., thin films, or fabrics. Expressed in Ω/sq or $\Omega/\square$; spoken „ohms-per-square“

How to measure $\Omega/\square$
(example): Make two copper probes that are 1 cm^2 (one cm on a side) and place them 1cm apart on the surface. Measure the resistance between them.



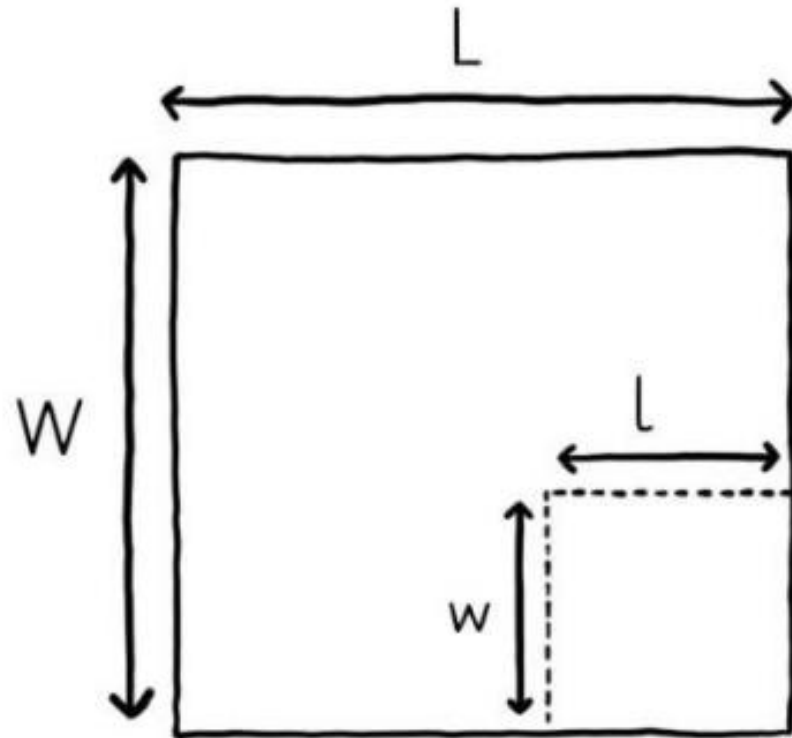
Surface Resistivity (Ω/sq or $\Omega/\square$)

Important: surface resistivity is dimensionless, i.e., it does not change as long as the measurement is performed correctly using a probe with square dimensions. Great for making comparisons!

$$p_s = \frac{p_v}{t} = \frac{W}{L} * R$$

if $L = W$; $l = w$:

$$p_s = R [\Omega/\square]$$



p_s = Surface Resistivity ($\Omega/\square$)

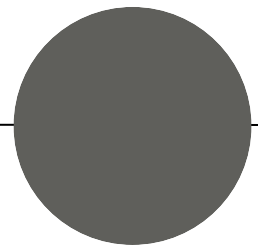
p_v = Volume Resistivity (Ω/cm)

R = Resistance (Ω)

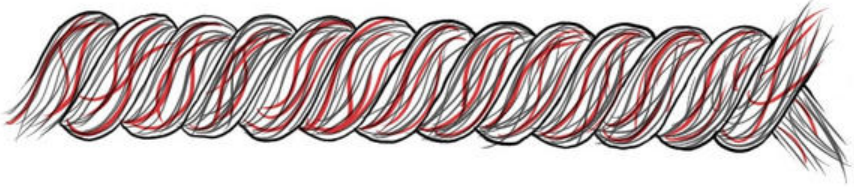
W = Width of electrodes

L = Distance between the probe electrodes (cm).

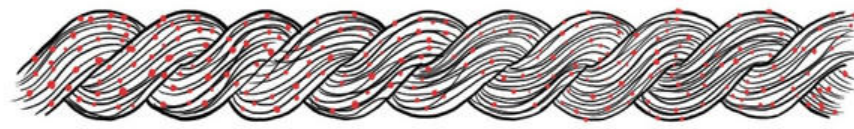
Conductive Threads, Yarns and Fibers



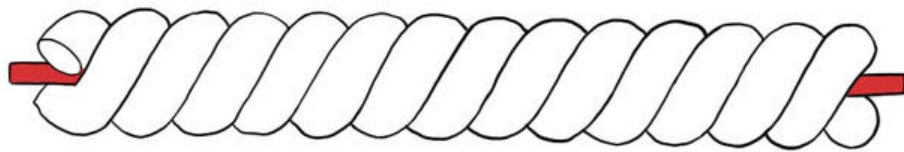
Different Types of Conductive Yarns



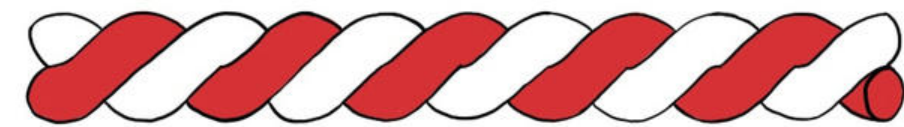
Spun from a mix of conductive and non-conductive filaments



Spun from non-conductive fibers or filaments, then coated in a thin layer of metal (metallized) or conductive polymer (polymerized)



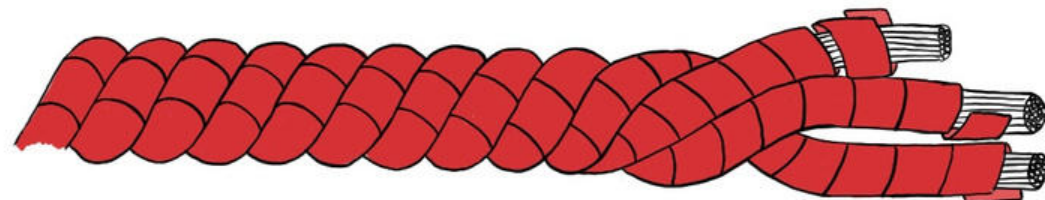
Metal-filled yarn (e.g., containing a stainless steel core)



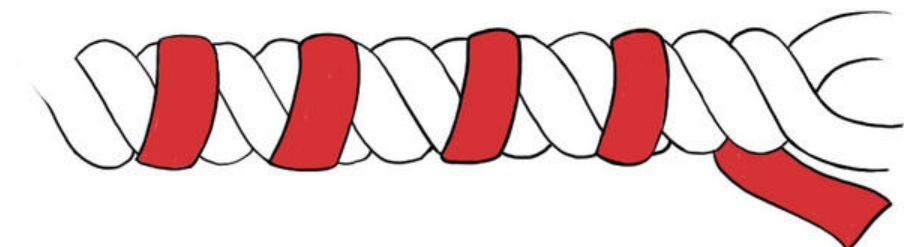
Conductive and non-conductive yarns plied together



Metal wrapped yarn



Filaments (e.g., nylon) wrapped in metal, then spun together



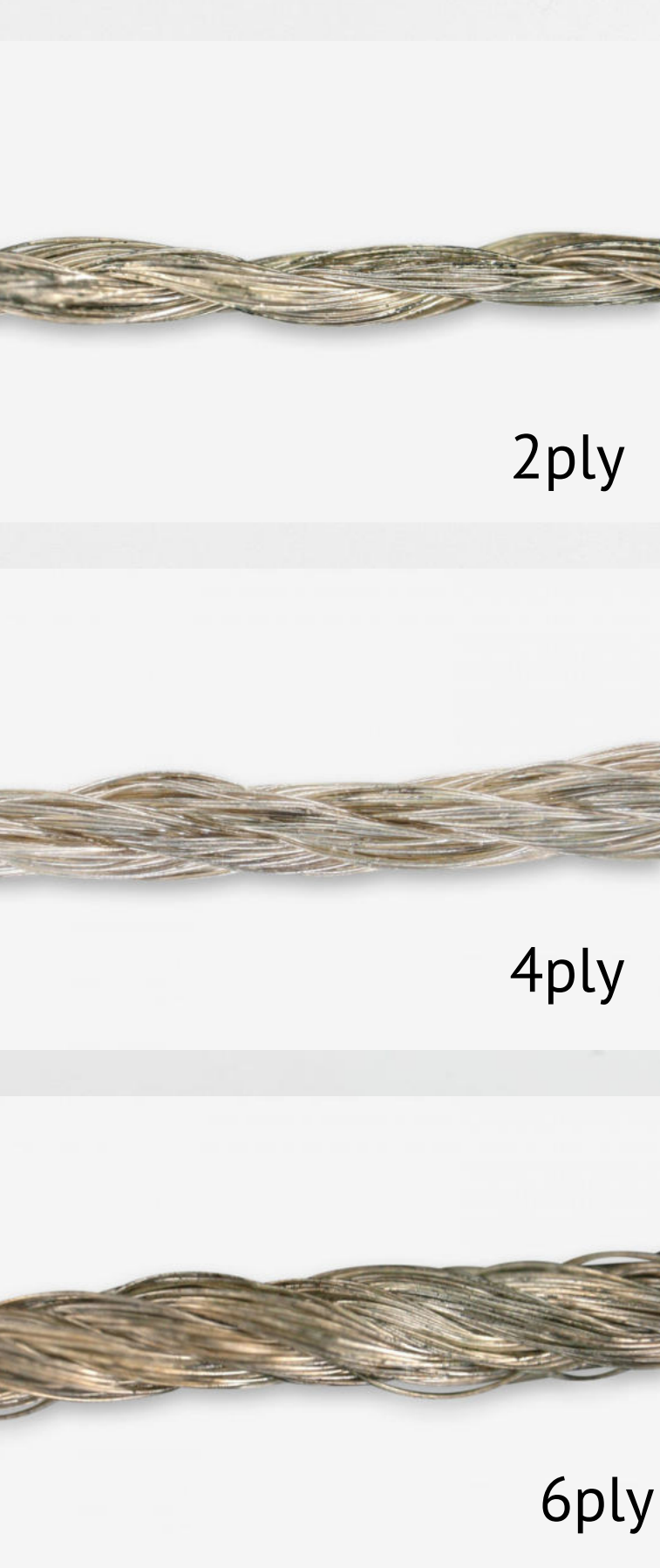
Spun fibers wrapped in metal (e.g., so-called gimp yarn)



Stainless Steel Thread

1. Stainless steel fibers spun together
2. Thickness and no of ply can vary, commonly found are 2ply, 3ply and (sometimes) 4ply-
3. Resistance: ca. 8.5 ohm/m (28 ohm/ft), i.e., **moderately conductive.**
4. Not (well) machine sewable (!), not solderable, rather “hairy” but can be waxed to be easier to sew.
5. Relatively wide-spread, e.g.,
 - a. <https://www.sparkfun.com/products/11791>
 - b. <https://www.sparkfun.com/products/13814>

Example Material



Metal-plated Thread

1. Silver-plated Nylon, optionally coated with e.g., Nitrile Rubber
2. Resistance: between 30 ohm/m and 80 ohm/m depending on ply, i.e., **moderately conductive.**
3. Smooth, machine sewable, not (!) solderable.
4. Tarnishes easily due to the silver (esp. when sewn by hand), coating may help.
5. Available as, e.g.,
 - a. Shieldex 235/36 x6 HCB (6ply, <30 Ω /m)
 - b. Shieldex 235/36 x4 HCB (4ply, <40 Ω /m)
 - c. Shieldex 235/36 x2 HCB (2ply, <80 Ω /m)
 - d. Sparkfun Conductive Thread 117/17 2ply. (legacy, still available in the lab)
 - e. Sparkfun Conductive Thread 234/34 4ply. (legacy, still available in the lab)



Metal-wrapped Nylon Thread

1. Thin flattened wires wrapped around a nylon core, then spun together (so called “Kupfer- or Silbergespinst, Lahnspinnung”)
2. Resistance: $> 2 \text{ Ohm/m}$ – i.e., **highly conductive**, e.g., for embroidered speakers
3. Still very flexible but slightly stiffer than metallized yarns, machine sewable.
4. Solderable!
5. Available as (e.g.) “Karl Grimm High Flex Kupfer 7x1” to business customers in larg(er) quantities
6. More information:
<https://www.kobakant.at/DIY/?p=379>



“Data Stretch”

1. Thin, flat copper wire wound as spiral around a flexible core, then wrapped with a stretchable textile cover.
2. Resistance: 5 Ohm/m – i.e., **highly conductive**, suitable e.g., for multitouch sensing.
3. Insulated due to the textile outer layer
4. Stretchable, about 30% elongation.
5. Requires a lot of practice to solder.
6. Available as (e.g.)
 - a. TibTech Data Stretch



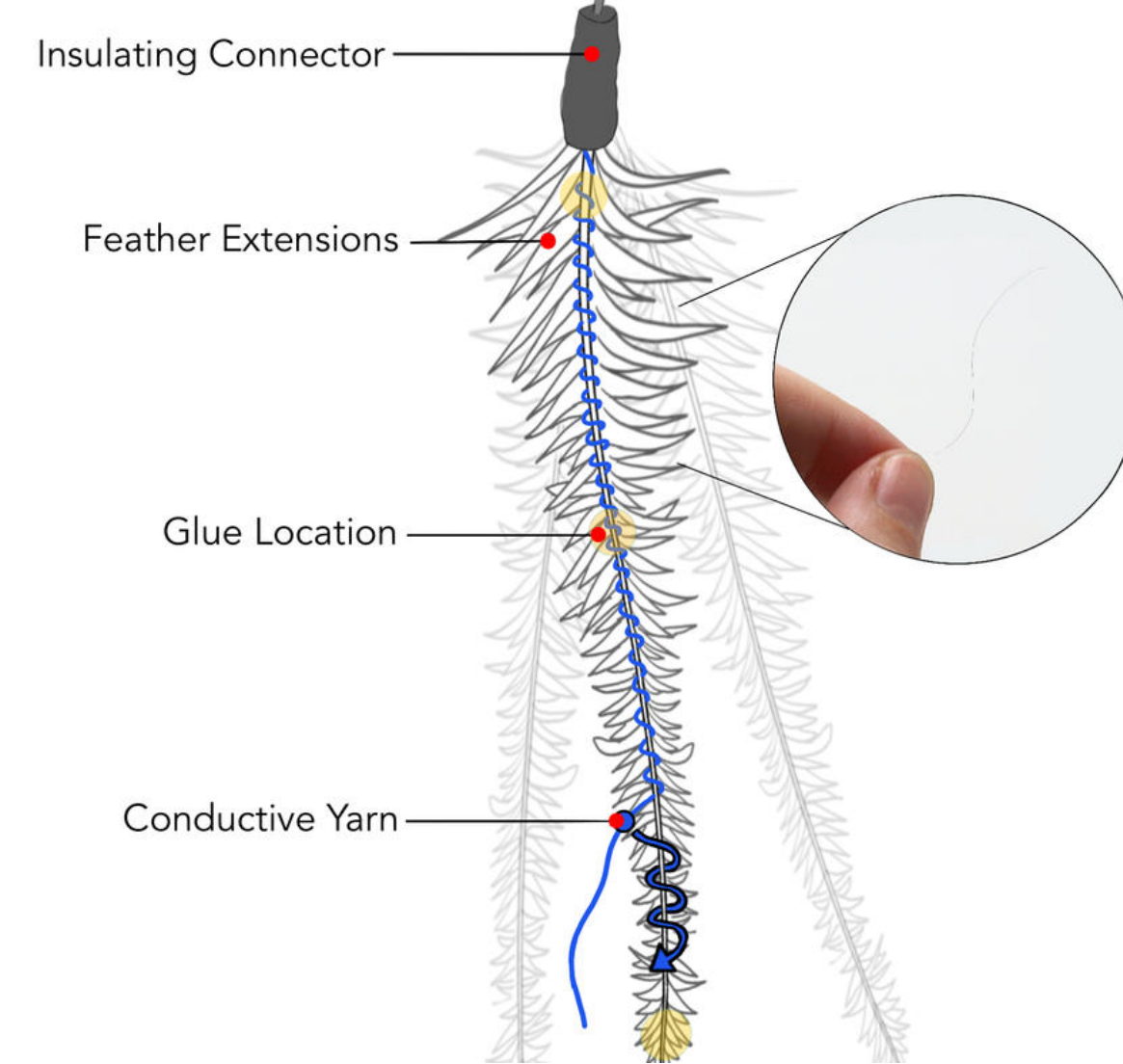
Conductive Wool

1. Conductive fibers (e.g., stainless steel) mixed in with (e.g.) merino wool.
2. Usually **moderately conductive**, can be shaped into sensors with **variable resistance**: movement changes resistance between fibers.
3. Can be used as a basis for felted sensors (e.g.,) or hand-spun into yarn and then knit, crocheted or braided.
4. Not solderable.
5. Available e.g., as
 - a. Bekinox by Baekert
 - b. Barts & Francis Conductive Wool



Metal Filaments

1. Pure metal (e.g., stainless steel), extruded extremely thin (e.g., $<0.04\text{mm}$)
2. Usually **highly conductive**.
3. Often used as a basis for compound yarns or mixed with wool, but also available as plain filaments.
4. Extremely thin, thus hard to handle, sew or solder.
5. Available e.g., as
 - a. <https://www.adafruit.com/product/1088>
 - b. [Barts & Francis 0.355mm Inox](#)
 - c. [Barts & Francis twisted Inox](#)



1. Featherhair (DIS 2022)

2. Marie Muelhaus, Jürgen Steimle, and Marion Koelle

3. <https://github.com/HCI-Lab-Saarland/FeatherHair>





Metal filled „Craft“

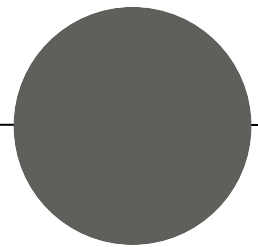
1. Fibers plied together or filled with a metal filament (often stainless steel).
2. Usually **highly conductive** but no data sheet (!)
3. Solderable but requires practice due to the filament's thin diameter.
4. Comes in a large variety of colors and materials, e.g., silk, cotton, merino, ...
5. Common brands/suppliers:
 - a. ITO
 - b. Habu
 - c. Barts & Francis

Jewelry Wire

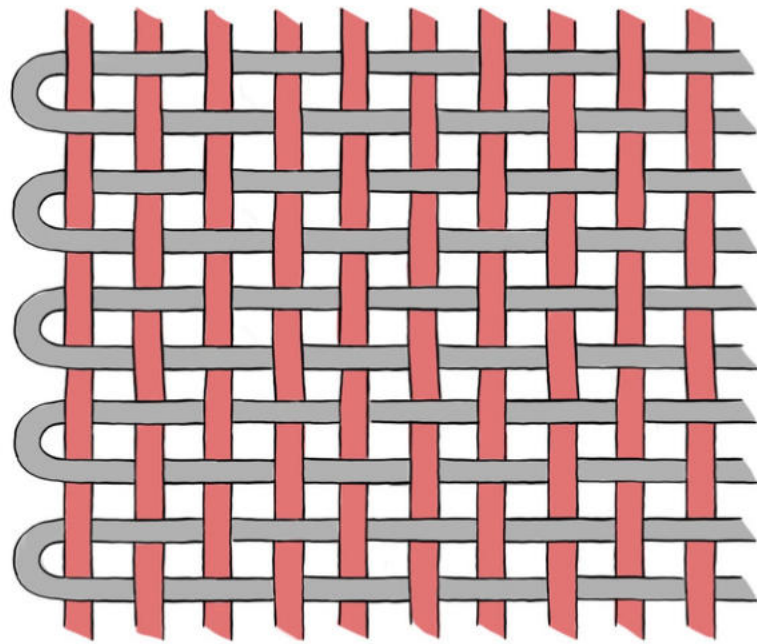
1. Often made from silver, copper, zinc or combinations thereof (e.g., brass).
2. Usually **highly conductive** but not data sheet (!)
3. Solderable (easy).
4. Can be blank or coated (e.g., with nylon).
5. Usually considerably thicker than metal filaments.
6. Rather stiff compared to conductive threads or yarns, not sewable
7. Easy to obtain, e.g., from craft supply stores.



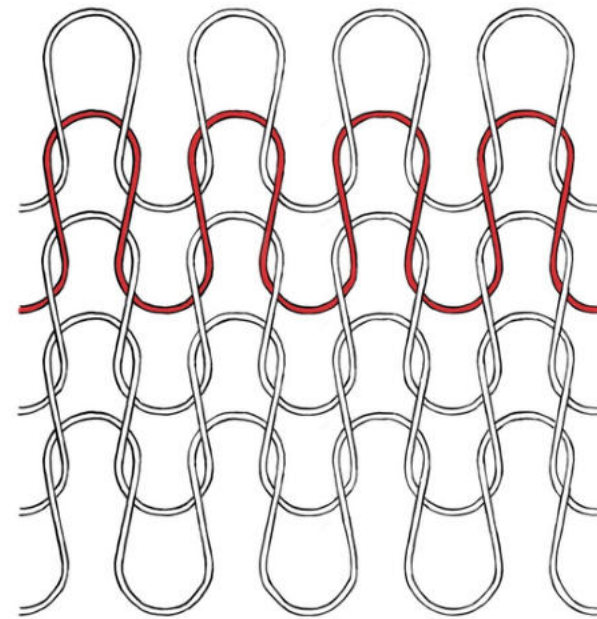
Conductive Fabrics



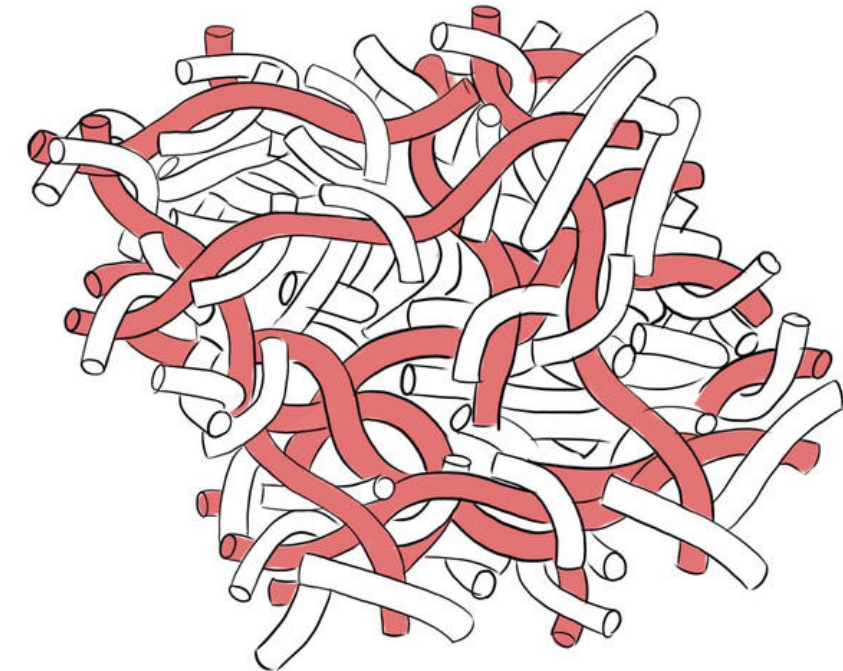
Types of (Conductive) Fabrics



1. **Woven**



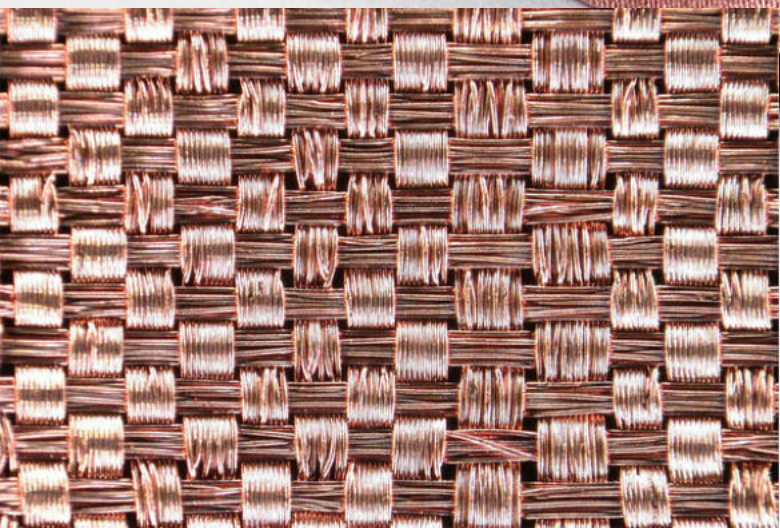
1. **Knit**



1. **Non-woven**

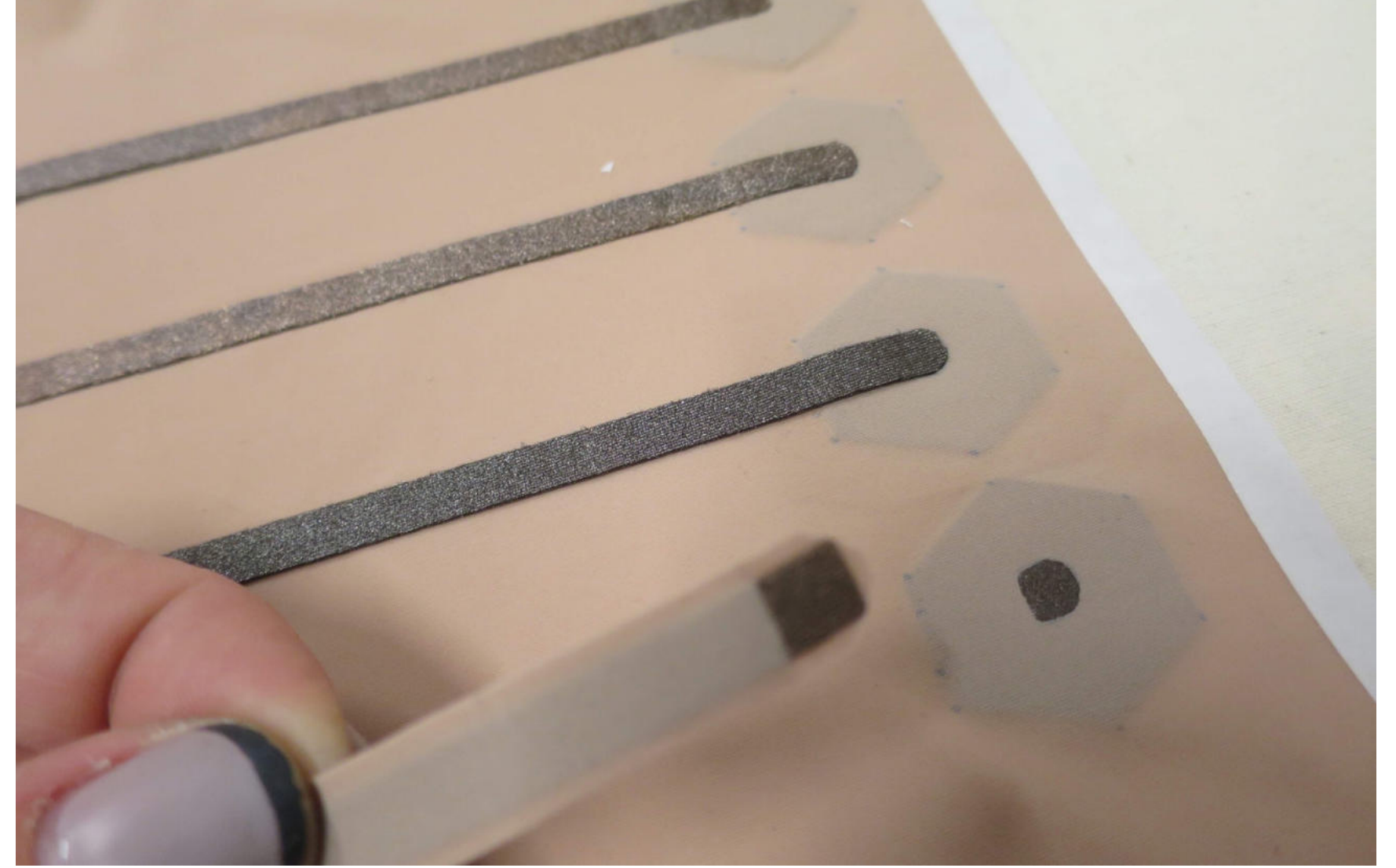
Metallized Nylon Fabric (Woven)

1. Metallized, woven textiles (e.g., nylon)
2. Different compositions of metals, e.g., copper, silver (Shieldex Kassel) or silver, copper and tin (Shieldex Zell).
3. Not stretchable
4. Sewable, can be ironed, some solderable
5. Surface resistivity (constant): $<0.02 - 0.03 \Omega/\square$, i.e., **highly conductive**.
6. **Important!** Conductive fabrics are also commonly used for shielding applications. Thus, some may contain nickel. **Do not use fabrics containing nickel** for wearable computing, especially not on skin!



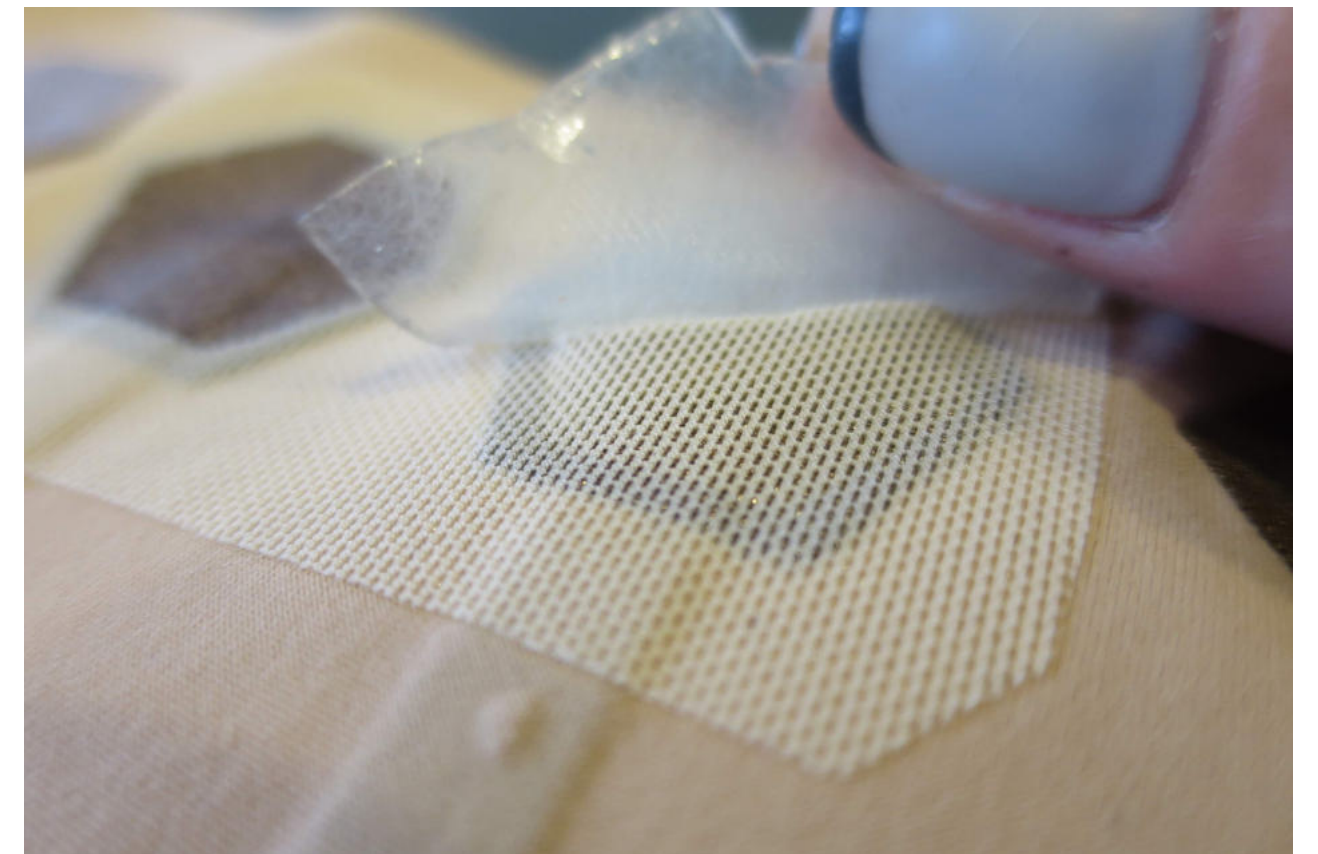
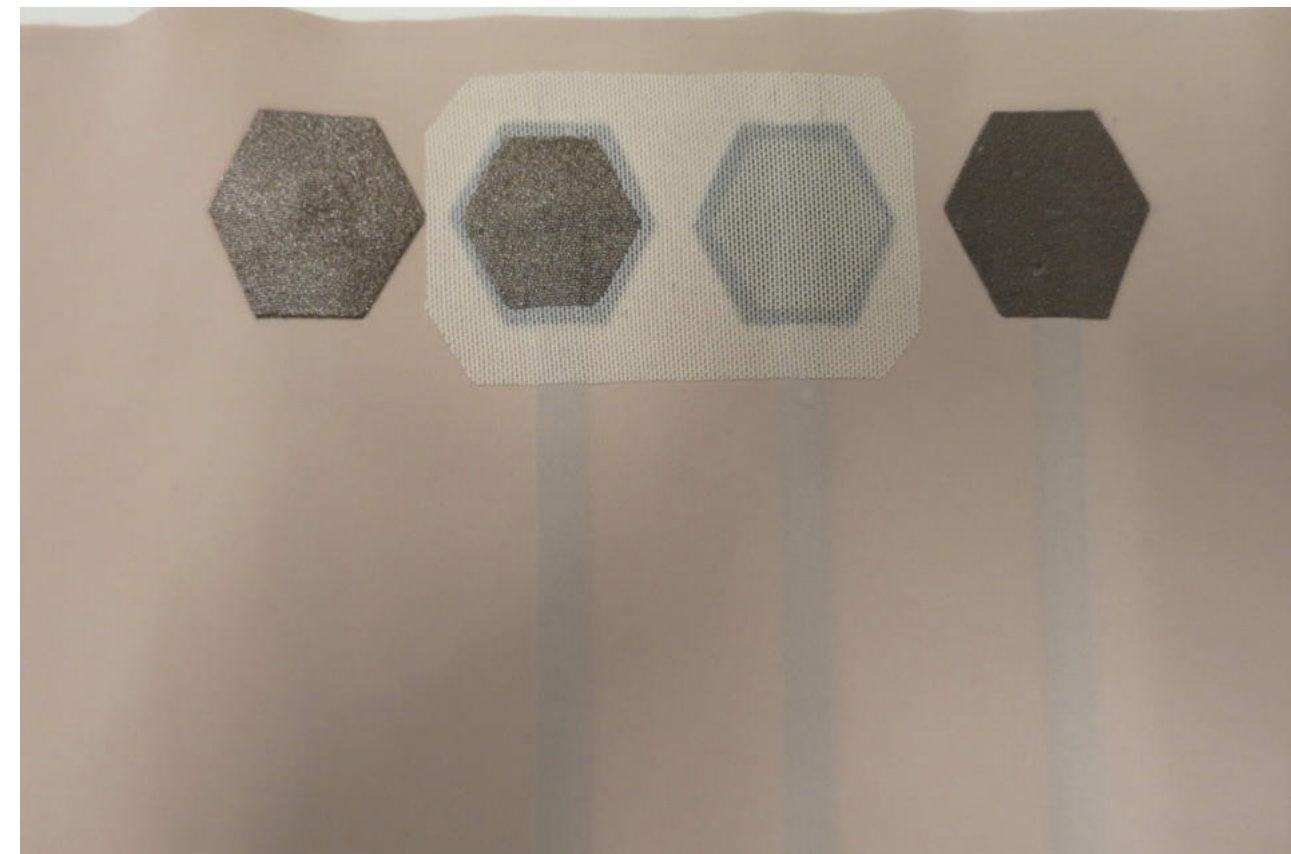
Stretchable (Knit) Conductive Fabric

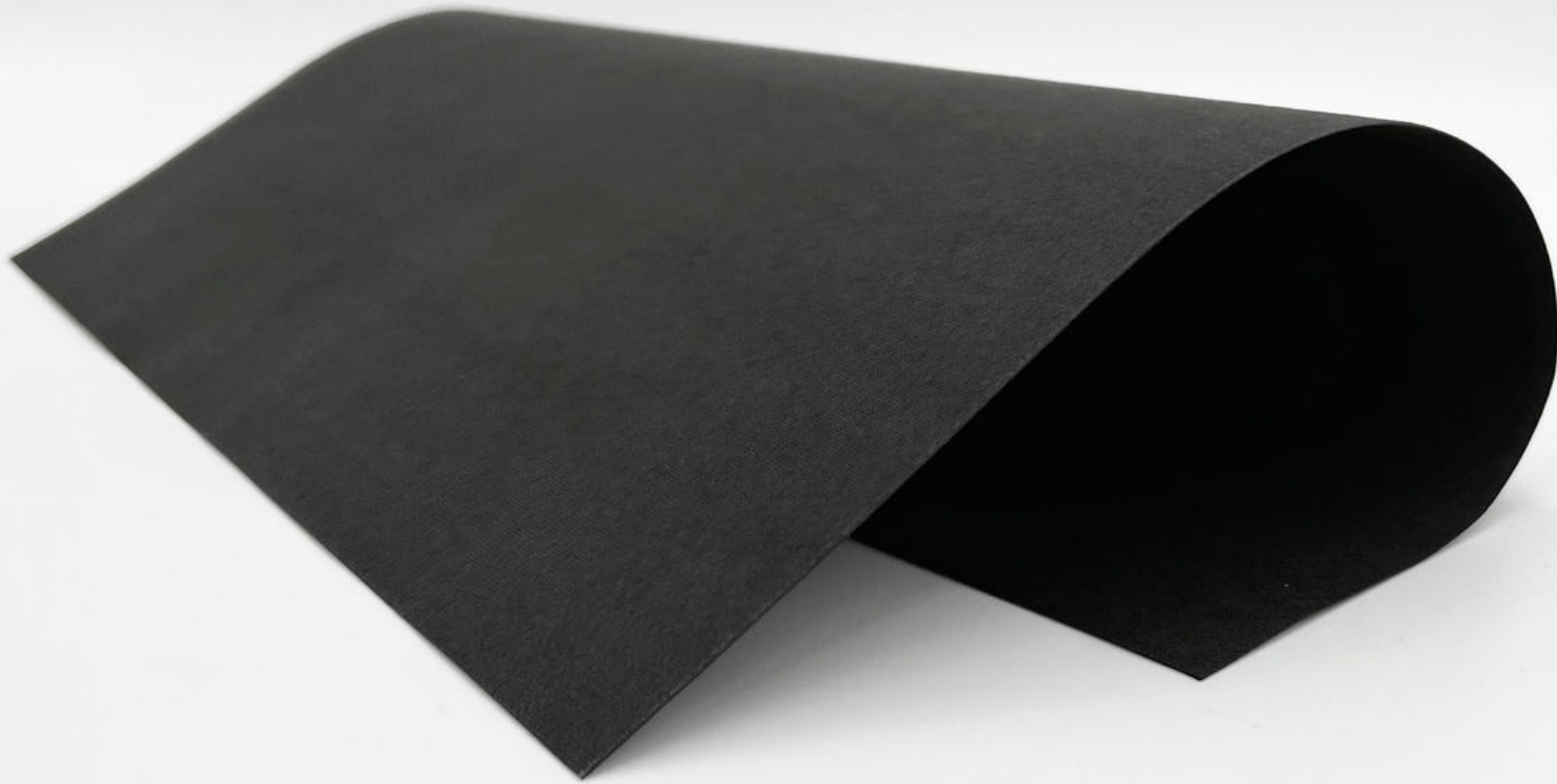
1. Metallized, knitted textiles (e.g., polyamide + elastomers)
2. Different variants, e.g., uni-directional or bi-directional stretch.
3. Sheet resistivity: can be $< 2 \Omega/\square$ (Technik-tex P130) to $< 5 \Omega/\square$ (Silitex), i.e., **highly conductive**, suited for on-skin electrodes.
4. Resistivity slightly variable, can be used as stretch sensors
5. Available as (e.g.,)
 - a. Shieldex® Technik-tex P130
 - b. Shieldex® Technik-tex P180
 - c. Shieldex® Silitex



1. Skillsleeves (TEI 2021)

2. Jarrod Knibbe, Rachel Freire, Marion Koelle, and Paul Strohmeier

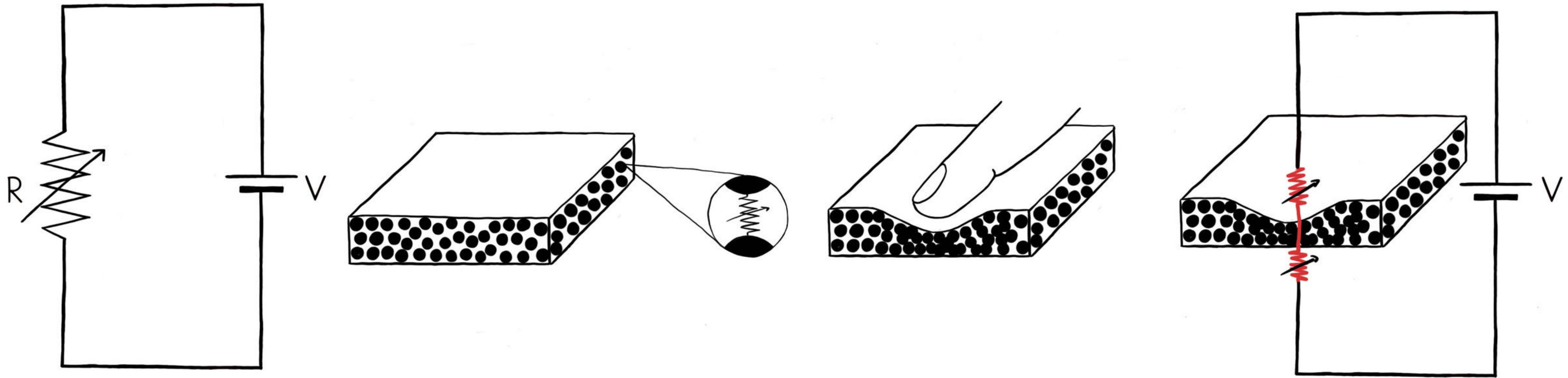


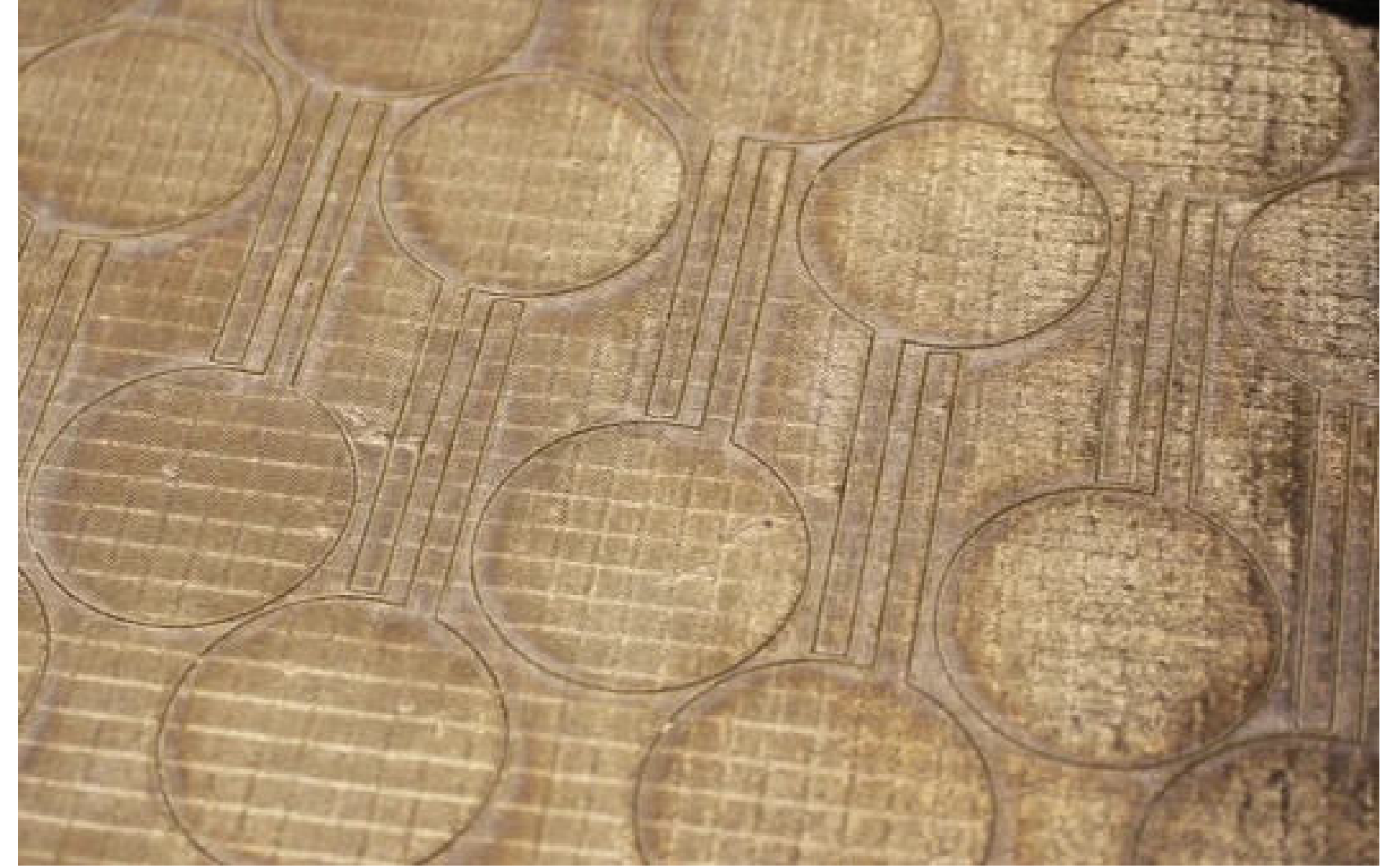


(Piezo) Resistive Fabric

1. Non-woven fiber, coated with conducted polymer (polymerized): "carbon-doped"
2. Relatively stiff and not stretchable
3. **Variable surface resistivity** of $8 \Omega/\square$ (compressed) to $105 \Omega/\square$ (relaxed)
4. High resistivity in relaxed state, suitable as heater fabric
5. Available, e.g., from Eeonnix:
 - a. EeonTex NW-170 PL-20
 - b. <https://www.adafruit.com/product/3670> (on Mouser)

Piezoresistivity



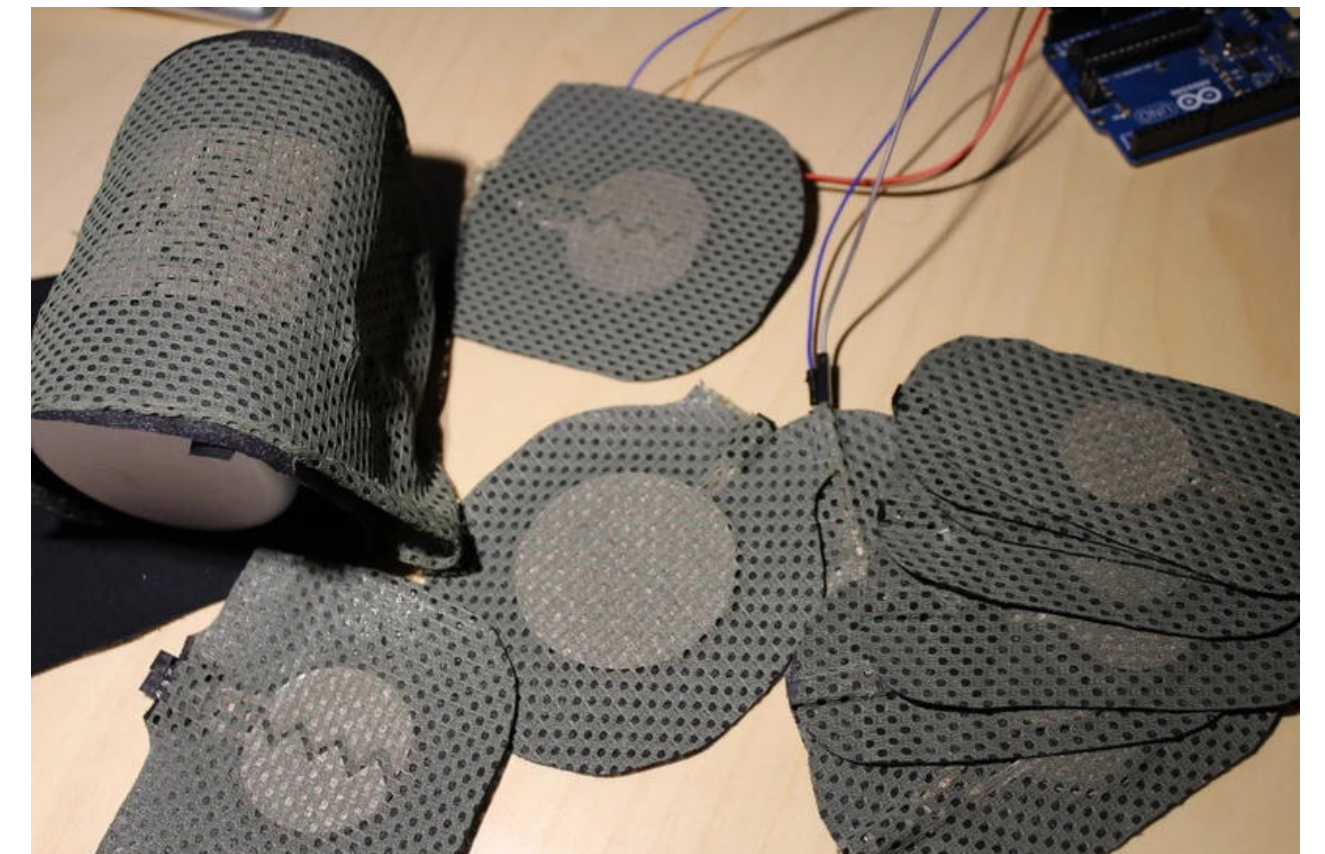


1. zPatch (TEI 2018)

2. Paul Strohmeier, Jarrod Knibbe, Sebastian Boring, and Kasper Hornbæk.

3. <https://zpatch.github.io>

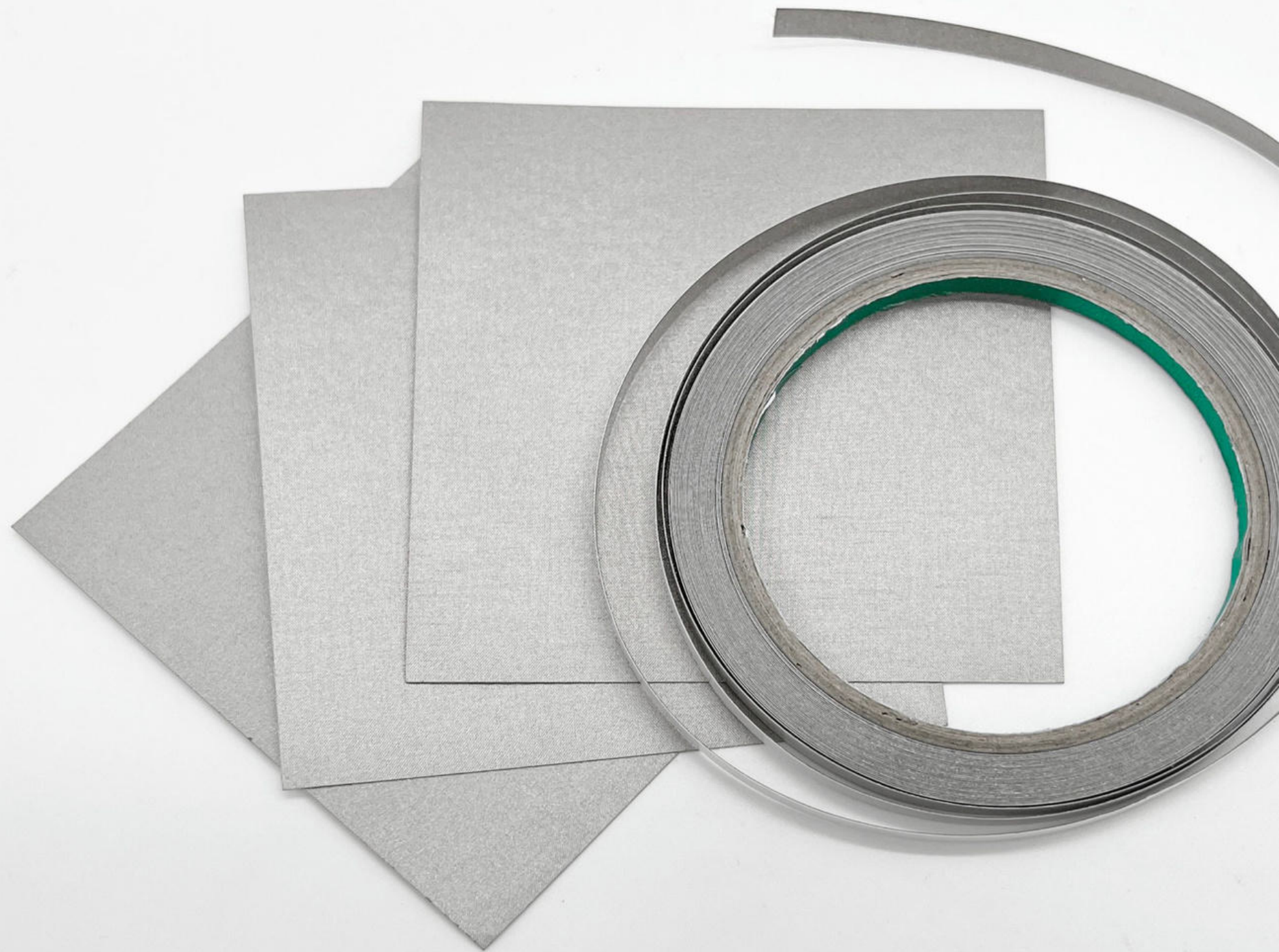
4. <https://www.instructables.com/ZPatch-Hybrid-ResistiveCapacitive-ETextile-Input/>





Resistive Sheet Material

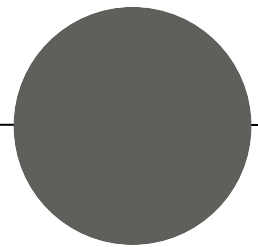
1. Polymer containing carbon particles "carbon-doped" cast into sheets
2. Usually sold as anti-static shielding material, re-purposed for e-Textile pressure sensing.
3. Relatively stiff and not stretchable
4. Heat sealable (e.g., using a soldering iron)
5. **Surface resistivity:** ca. $31 \Omega/\square$
6. Available, e.g., as "Velostat" or "Linqstat":
 - a. <https://www.adafruit.com/product/1361> (on [Mouser](#))

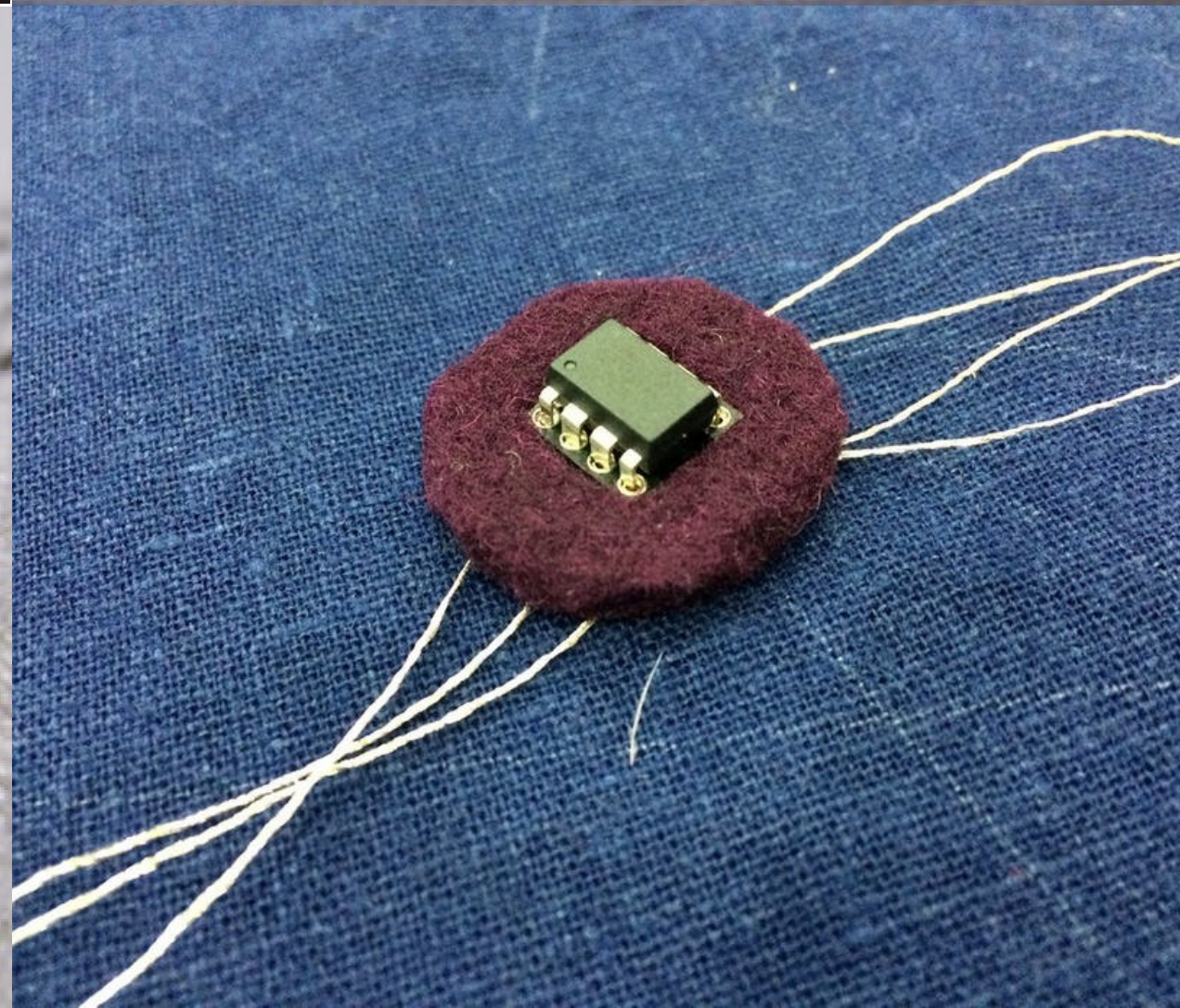
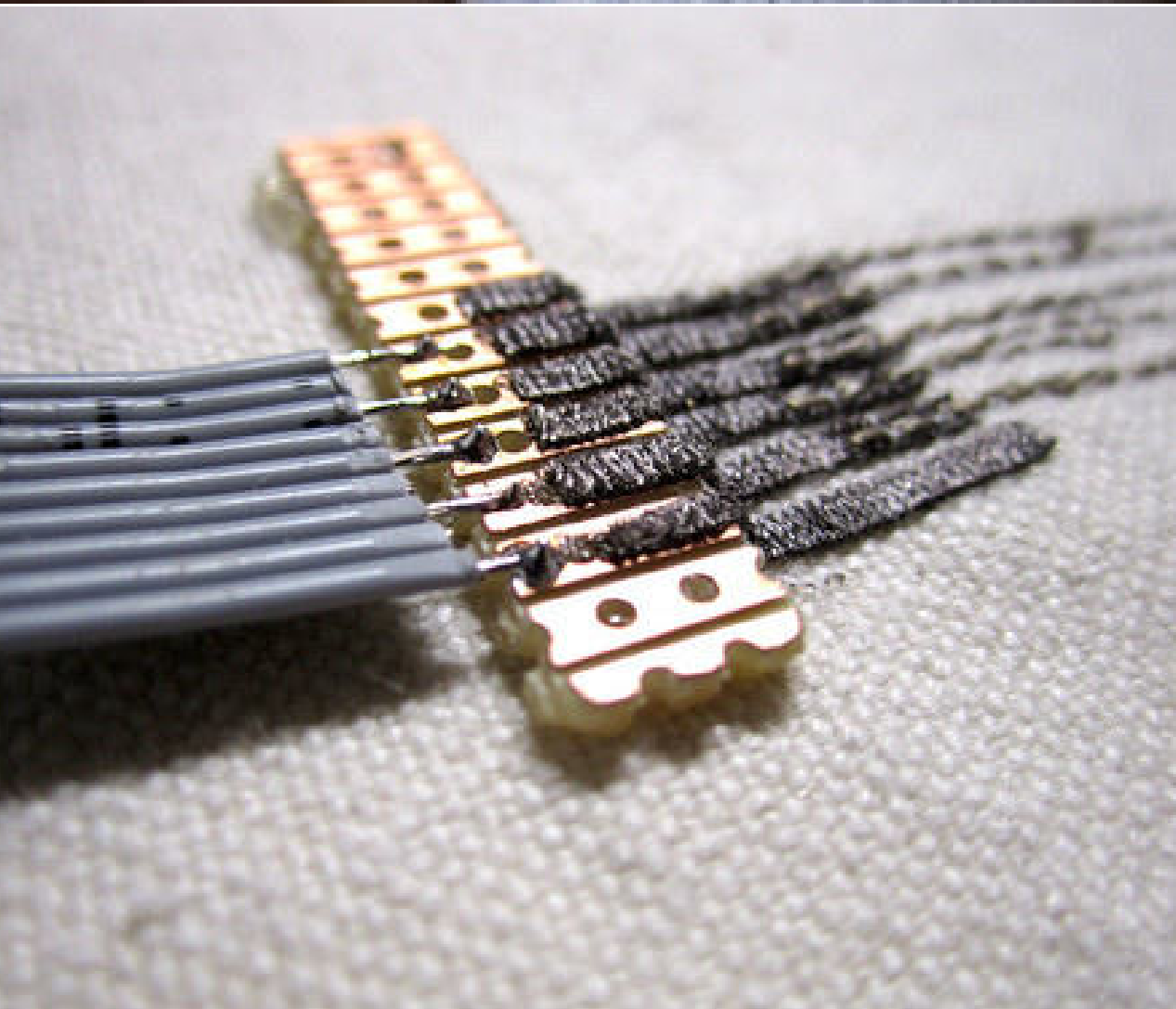
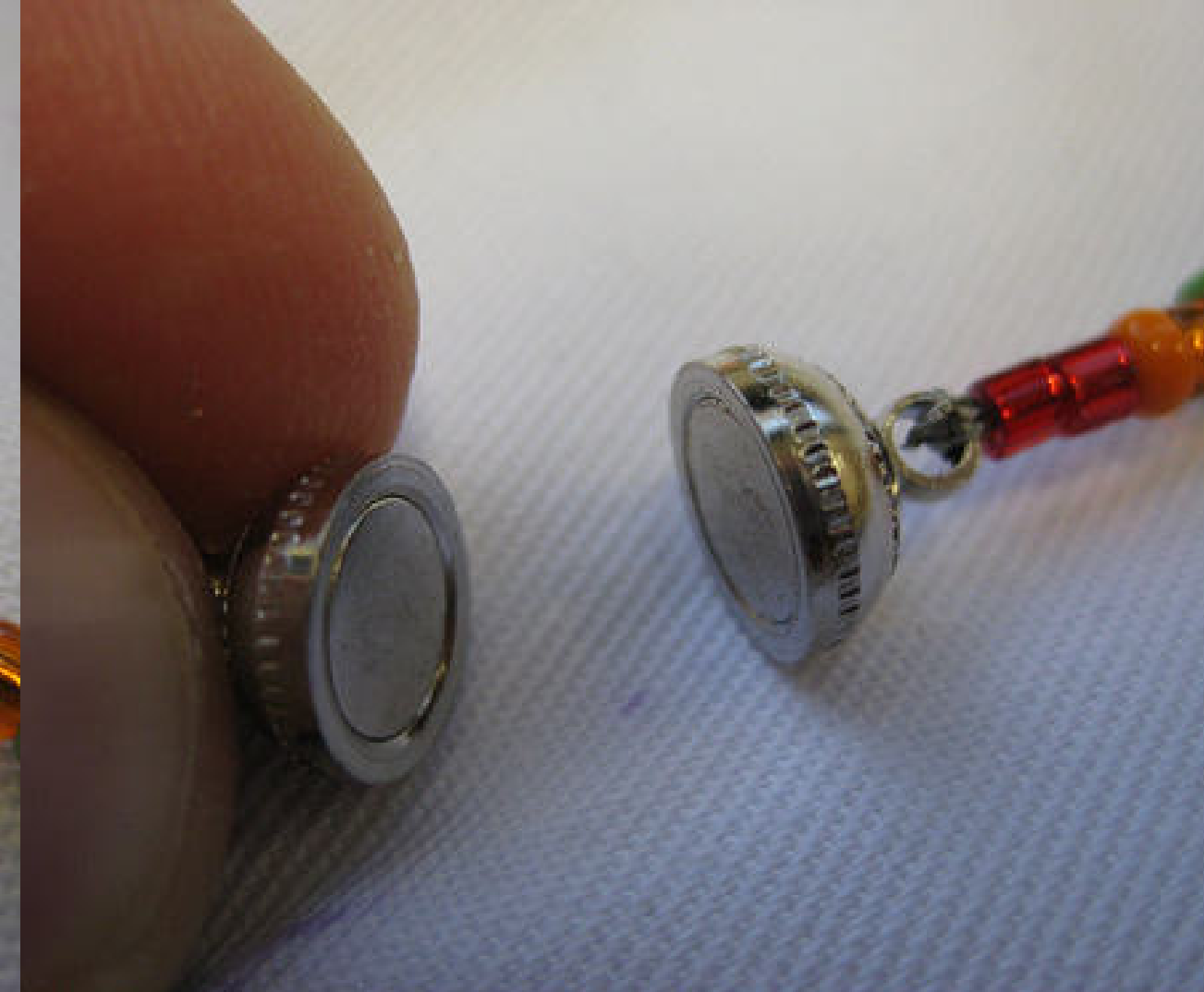


Silver-plated Nylon Adhesive Fabric

1. Metallized silver nylon fabric (or tape) with adhesive backing.
2. Relatively stiff compared to other conductive fabrics.
3. Easy to handle, easy to cut into shape.
4. Surface resistivity (constant): approx. 5-6 $\Omega/\square$., i.e., **highly conductive**
5. Solderable (with practice and flux)
6. Easily available, e.g., from Adafruit or Mouser:
 - a. <https://www.adafruit.com/product/4762> (on Mouser)
 - b. <https://www.adafruit.com/product/3960> (on Mouser)

Component Connection





Hannah Perner-Wilson and Mika Satomi: Kobakant,
Examples online at:

- <https://www.kobakant.at/DIY/?p=1272>

and

- <https://www.kobakant.at/DIY/?p=7963>

E-Textile Cheat Sheet CONNECTIONS v0_2017									
	conductive thread (solderable)	conductive fabric	copper tape	flex pcb	solid core wire multistranded wire	pcb	pins, leads	smd pads	snap fasteners
conductive thread (solderable)									
conductive fabric									
copper tape									
flex pcb									
solid core wire multistranded wire									
pcb									
pins, leads									
smd pads									
snap fasteners									

E-Textile Connection Cheat Sheet

1. Challenges:
2. electrical connections (i.e., „making contact“)
3. insulating overlapping traces/components (i.e., “preventing contact“)
4. different techniques, inspired from electronics and crafts > cheat sheet

Meet the Materials | Group Work

Material (e.g., yarn, fabric)	Method & Process	Measured R	The Material's Resistivity

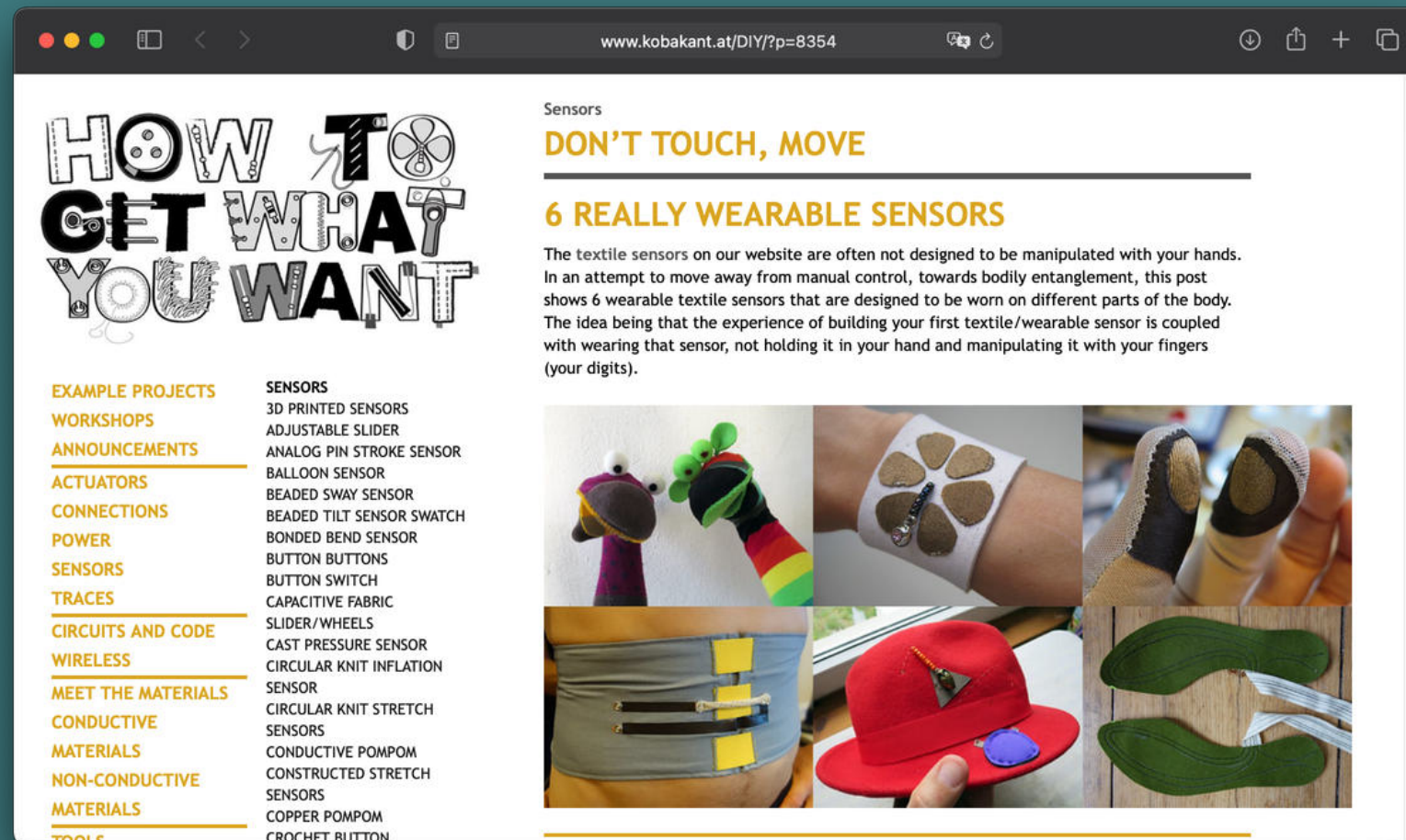


Your Task: Pick a multimeter and explore the available materials.

- 1. Test a selection of the materials from the material collection.
- 2. Determine **surface resistivity** (for thin, sheet material) or **volume resistivity** (for yarns or foam material). Does your measure equal what is provided on the data sheet (if available)?
- 3. Prepare to **discuss your results** orally in class (max. 5 Min). Use above table as orientation.

(max. 15 Min)

Literature / Further Reading

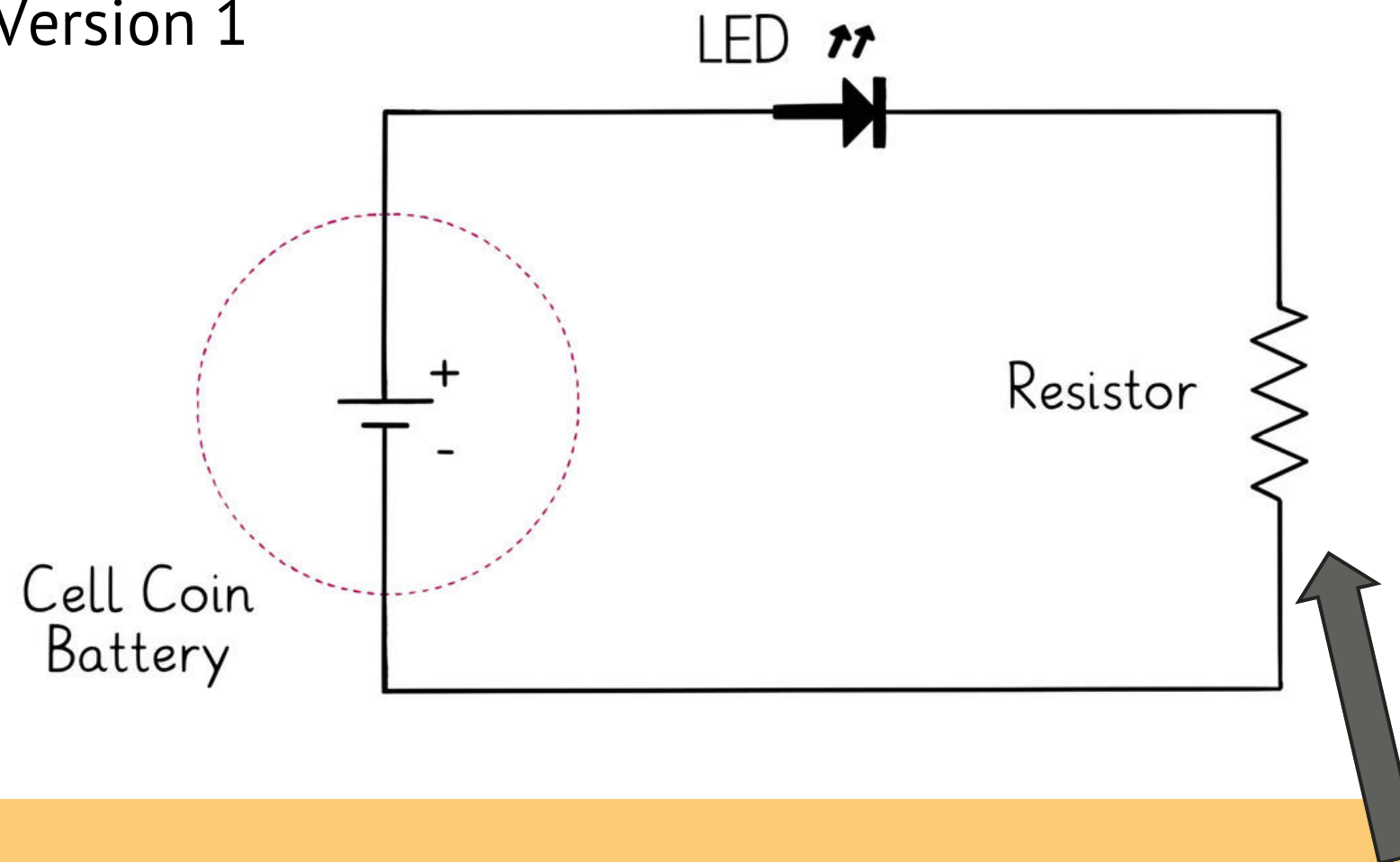


1. Perner-Wilson, H., Satomi, M.: Kobakant (www.kobakant.at), E-Textile Knowledge Repository, Material Data Base & Blog, e.g., “Bend Sensor Glove”
2. Perner-Wilson, H., Buechley, L., & Satomi, M. (2010, January). Handcrafting textile interfaces from a kit-of-no-parts. In *Proceedings of the fifth international conference on Tangible, embedded, and embodied interaction* (pp. 61-68). see also: <http://konp.plusea.at>
3. Buechley, L., Eisenberg, M., Catchen, J., & Crockett, A. (2008, April). The LilyPad Arduino: using computational textiles to investigate engagement, aesthetics, and diversity in computer science education. In *Proceedings of the SIGCHI conference on Human factors in computing systems* (pp. 423-432).

E-Textile Exercise

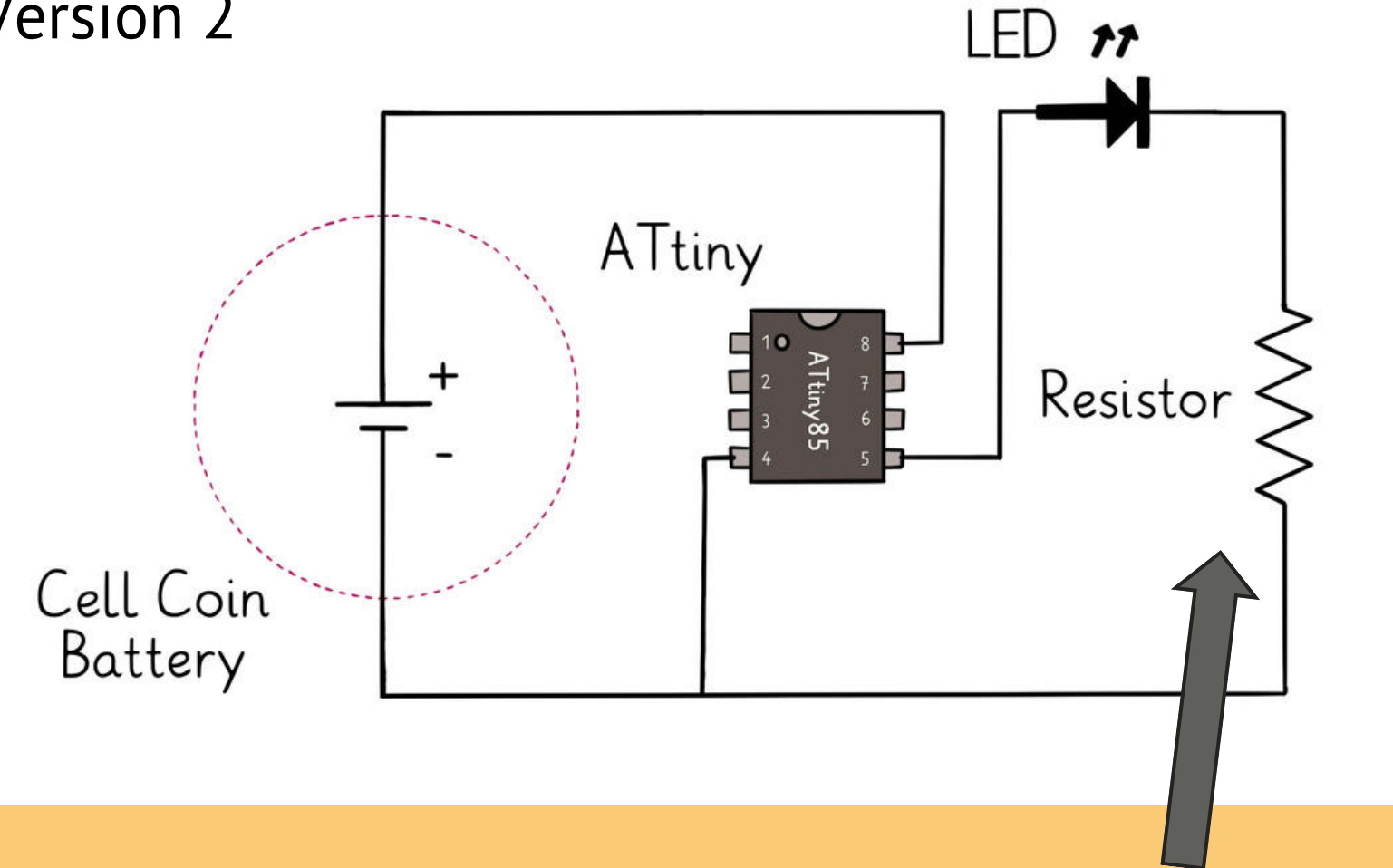
Version 1 is mandatory; Version 2 is a bonus exercise. Pick based on your skill level/previous knowledge and time commitment!

Version 1



OR

Version 2



The LEDs we're using already have a resistor built in, so they can be connected directly to the positive and negative terminals of the battery.